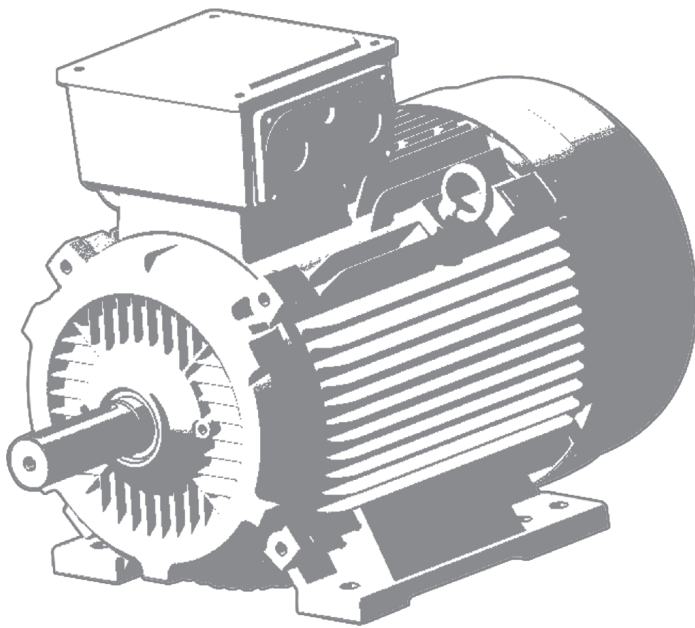


Standard Motor Catalogue

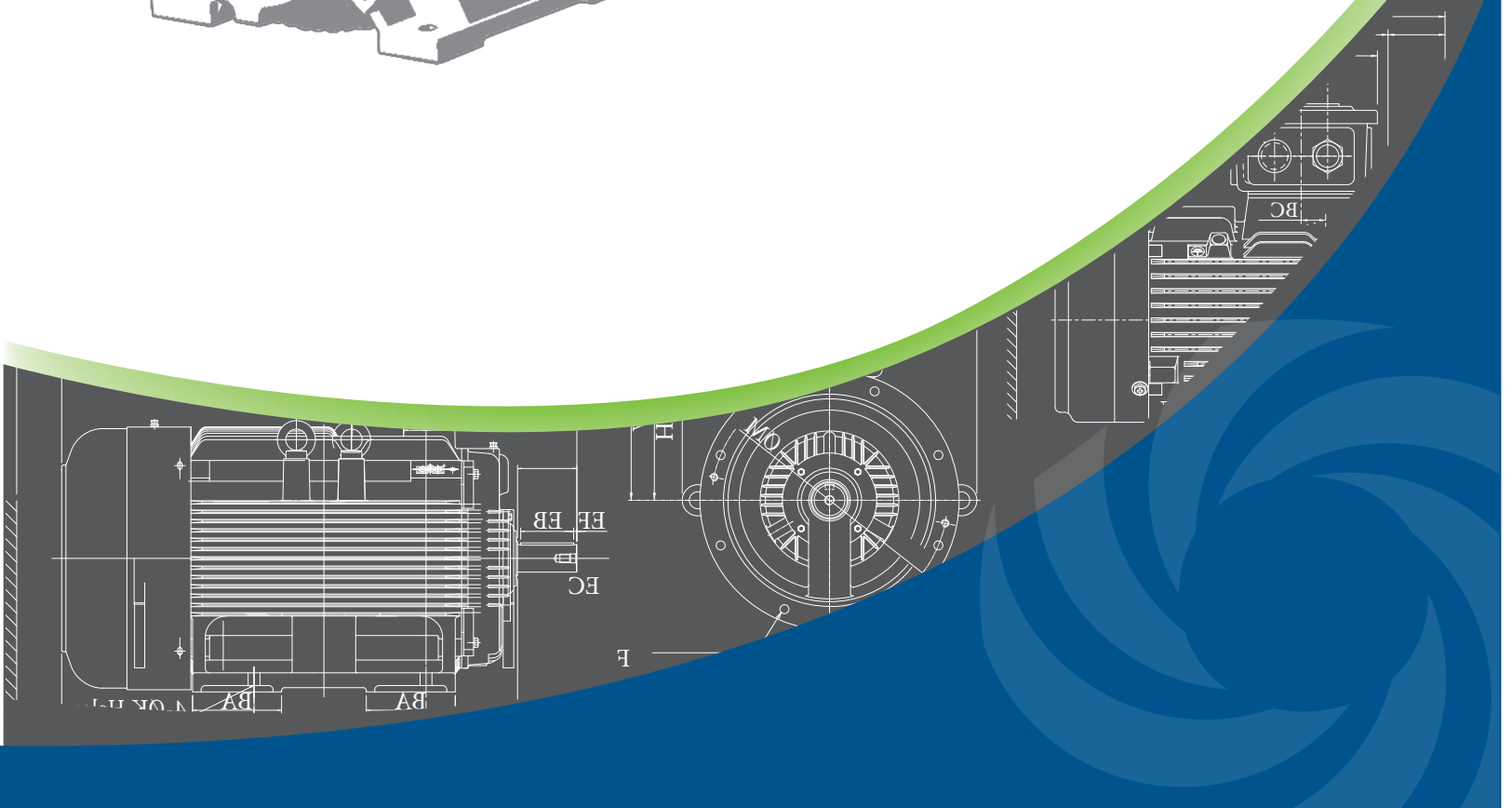
AESV2E / AESU2E Series

AESV3E / AESU3E Series



TECO
e-motion

IE2 IE3





TECO Group

Growth-Challenge-Unlimited Exploration

TECO, a famous globalize enterprise group, share the third in global industrial motor of the world; and it has successfully diversified into a highly competitive development conglomerate with worldwide business operations including heavy electrical, home appliance, electronics, IT system, telecommunications equipments, financial investment from the motor giant enterprises. TECO Group currently has nearly hundred oversea affiliates and subsidiaries, and the total global employee amount is over tens of thousands. It has a wide range technology cooperation with many world famous enterprise, such as GE, Japan Yaskawas, the U.S. Westinghouse, Siemens of Germany, Japan, Hitachi, Mitsubishi and so on, its business territory has been extended to Asia, America, Europe, Australia, and became known as a well-known World Group, TECO had become an international brand.

TECO History

- 1956** Company foundation and start of production in the first TECO factory in San-Chung, Taiwan
- 1965** Production capacity increased by opening new factory in Shin-Chuan, Taiwan
- 1966** Start of close technical co-operation with Hitachi, Yaskawa and Taiyo.
- 1979** Further production capacity increase by opening a factory in Chung Li, Taiwan Plant I (Heavy Motor Plant)
- 1987** Opened Chung Li Plant II for serial motor production (Small Motor Plant)
- 1987** Started Joint venture with Westinghouse Motor Company, USA, one of the leading motor manufacturers in North America
- 1991** Foundation of TECO Perai, Penang Province, Malaysia, to serve the local market with low voltage motors
- 1995** 100% take over of Westinghouse motor business by TECO
- 1995** Foundation of TECO Westinghouse Motor Company
- 2000** Opened TECO factory for low voltage motors in Suzhou, Jiangsu Province, China
- 2003** Opened TECO factory for low and medium voltage motors in Wuxi, China
- 2005** Opened third TECO factory in Nanchang, Jjiangxi Province, China
- 2006** Opened factory in Huyen Long Thanh, Tinh dong Nai Province, Vietnam to increase local business
- 2008** Opened TECO repair and assembling plant in Dammam, Kingdom of Saudi Arabia, on joint venture basis
- 2008** Established TECO fuan in Fujian Province, China, with a decision for construction of a new plant
- 2010** Starting the prodcution of aluminum motors and semi finished aluminum parts at TECO Fuan factory in Fujian Province, China.
- 2012** Established India Branch Company.
- 2013** Established Turkey Branch Company.
- 2014** Established bonded warehouse in Netherlands.



Performance Characteristics

Max. r.p.m

Safe running speed for the squirrel-cage induction motors. Unless otherwise the name plate specifies, all the squirrel-cage, 3 phase, induction motors lower than 1000V and smaller than 315 Frame Size can safely run continuously at the speed in the table below.

The safe Max. r.p.m. for the squirrel-cage, 3 phase induction motors of 1000V or lower.

Frame Size	2 Pole	4 Pole	6 Pole
≤ 100	5200	3600	2400
112	5200	3600	2400
132	4500	2700	2400
160	4500	2700	2400
180	4500	2700	2400
200	4500	2300	1800
225	3600	2300	1800
250	3600	2300	1800
280	3600	2300	1800
315	3600	2300	1800

Note: When the motors run above the rated speed, for example, using in speed controller, the noise and vibration will increase. In this situation, the motors are required to be corrected to satisfy the acceleration ability above the rated speed. Besides, the bearing lifetime will decrease. Pay attention to the time for adding the oil and grease to insure its lifetime.

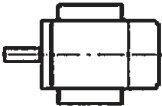
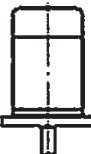
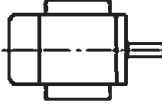
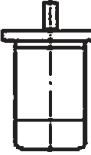
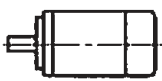
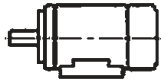
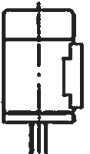
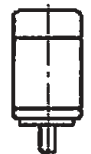
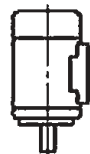
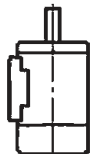
Grease Life (Horizontal installation)

Frame Size	Pole	Grease lifetime up to CT 40°C
Grease for permanent lubrication bearing		
80~225	2, 4, 6	20000 hours (B3 mounting)
Grease for regreasable bearing		
250~280	2	3000 hours
250~280	4, 6	8000 hours
315~355	2	2000 hours
315~355	4, 6	4000 hours

Note:

1. If the coolant temperature is increase by 10K, the grease lifetime and regreasing interval are halved.
2. 2000 hours apply to horizontally installed motors with coupling transmission.

Mounting Arrangement (IM code)

Foot Mounted		Flange Mounted		Foot / Flange Mounted	
IM 1001 (IM B3)		IM 3001 (IM B5)		IM 2001 (IM B35)	
Horizontal Shaft. Foot mounted.		Horizontal Shaft. 'D' type flange at D.E. No feet.		Horizontal Shaft. 'D' type flange at D.E. Foot mounted.	
IM 1051 (IM B6)		IM 3011 (IM V1)		IM 2011 (IM V15)	
Horizontal Shaft. Foot wall mounted with feet on left-side when viewed from D.E.		Vertical Shaft. 'D' type flange at D.E. Shaft down. No feet.		Vertical Shaft. 'D' type flange at D.E. Wall mounted. Shaft down.	
IM 1061 (IM B7)		IM 3031 (IM V3)		IM 2031 (IM V36)	
Horizontal Shaft. Foot wall mounted with feet on right-side when viewed from D.E.		Vertical Shaft. 'D' type flange at D.E. Shaft up. No feet.		Vertical Shaft. 'D' type flange at D.E. Wall mounted. Shaft up.	
IM 1071 (IM B8)		IM 3601 (IM B14)		IM 2101 (IM B34)	
Horizontal Shaft. Ceiling mounted with feet above motor.		Horizontal Shaft. 'C' type flange at D.E. No feet.		Horizontal Shaft. 'C' type flange at D.E. Foot mounted.	
IM 1011 (IM V5)		IM 3611 (IM V18)		IM 2111	
Vertical Shaft. Wall mounted. Shaft down.		Vertical Shaft. 'C' type flange at D.E. Shaft down. No feet.		Vertical Shaft. 'C' type flange at D.E. Wall mounted. Shaft down.	
IM 1031 (IM V6)		IM 3631 (IM V19)		IM 2131	
Vertical Shaft. Wall mounted. Shaft up.		Vertical Shaft. 'C' type flange at D.E. Shaft up. No feet.		Vertical Shaft. 'C' type flange at D.E. Wall mounted. Shaft up.	

It is important to nominate the "IM" code at enquiry and order stage to ensure that drain holes are in the correct position and bearing arrangement is checked for suitability if the "IM" code differs from standard.

Standard Mounting Arrangement

TECO Stock Motors - Standard Mounting Arrangement and Terminal Box Position				
Mounting	IM Code	(IM Code)	Terminal box position (viewed from drive end)	Cable entry direction
Foot	IM1001	IMB3	Right	From below
Flange	IM3011	IMV1	As needed (motor can be rotated)	From flange end
Foot & Flange	IM2001	IMB35	Right	From below

SPECIFICATION TABLE

IE2 EFFICIENCY 3-PHASE INDUCTION MOTORS

TYPE: AESV2E, AESU2E

ITEM		STANDARD SPECIFICATION
RATING	Kind of Motors	Squirrel - Cage Induction Motors (SCIM) .
	Design Standards	IEC 60034, IEC 60072 .
	Voltages	230V, 400V, 690V, 265V, 460V .
	Frequency	50Hz or 60Hz .
	Output Range	0.37 kW ~ 315 kW (50Hz) or 0.43 kW ~ 362 kW (60Hz) .
	R.P.M. (Syn.)	3000 ~ 750 R.P.M. (2 ~ 8 Poles) or 3600 ~ 900 R.P.M. (2 ~ 8 Poles) .
	Time Duty	Continuous. S1 , S.F. : 1.0 .
	Frame Nos.	80M ~ 315D .
	Protection Enclosure	Totally Enclosed (IP 55) .
	Cooling Method	Self External Fan, Surface Cooling (IC 411) .
	Mounting	Horizontal Foot Mounted B3 (IM 1001) . Horizontal Flange Mounted B5 (IM 3001) . Horizontal Foot And Flange Mounted B35 (IM 2001) . Vertical Flange Mounted , Shaft Down V1 (IM 3011) .
APPLICATION	Environment Conditions	Place : Shadow, Non-Hazardous. Ambient Temperature : -20 ~ 40°C . Relative Humidity : Less Than 90%RH (Non-Condensation) . Altitude : Less Than 1,000 Meters .
	Power Source Conditions	Voltage : $\pm 10\%$, Frequency : $\pm 5\%$, and 10% Max. of Combined Voltage and Frequency. But Frequency Variation Does Not Exceed $\pm 5\%$.
	Method of Starting	Full Voltage Direct On Line or Y - Δ Starting .
	Drive Method	Coupling Or Belt Servie Are Available For All Range Depends On Bearing Capacity. Bearing Capacity Refers To Radial Bearing Load .
	Direction of Rotation	CW According to IEC Definition, Suitable For Bi - Directional Operation .
APPLICATION	Bearing	Bracket Mounting , Vacuum De - Gassed High Quality Open Bearings for Frame Nos. 180M ~ 315D, Grease Pre - Packed Shielded Rolling Bearings for the Others .
	Lead Terminal	Solderless Lug Terminals .
	Terminal Box	Cast iron, Can be Rotated Each 90° With Conduit Hole for Cable Entrance. The Terminal Box is Usually Placed on The Top of The Frame and The Cables Could be Connected from Four Possible Directions .
	Stator Insulation	Class F Insulation System .
	Painting	Phenolic Rust Proof Base Plus Lacquer Surface Finished Painting in Pebble - Gray Color (Munsell 5Y 7.5/1) (RAL 7032) .
	Bolt Thread	ISO Metric System (Strength Catagory 8.8T) .
	Grounding Terminal	Two Terminals . One inside The Terminal Box, Another One On The Outside Of The Frame .
PERFORMANCE	Test Procedure	IEC 60034-2-1:2007 And Full Voltage Measuring Starting Performance .
	Winding Temperature Rise	Not to Exceed 80 Rise by Resistance Method at S.F 1.0 Operation .
	Over Speed	120% Syn. R.P.M. for 2 Min .
	Over Torque	160% Rated Torque for 15 Sec .

TEFC, CLASS F, 40°C AMBIENT TEMP., IEC DESIGN N CONTINUOUS DUTY, S.F. 1.0
400V/50HZ

OUTPUT		FULL LOAD rpm	FRAME NO.	EFFICIENCY				POWER FACTOR				CURRENT		TORQUE				ROTOR GD2 kg-m2	APPROX. WEIGHT kg
HP	KW			FULL LOAD (%)	3/4 LOAD (%)	2/4 LOAD (%)	1/4 LOAD (%)	FULL LOAD (%)	3/4 LOAD (%)	2/4 LOAD (%)	1/4 LOAD (%)	FULL LOAD (A)	LOCKED ROTOR (A)	FULL LOAD N•m	LOCKED ROTOR %FLT	PULL UP %FLT	BREAK DOWN %FLT		
0.5	0.37	915	80M	65.5	63.8	57.9	40.5	65.0	55.5	44.0	31.0	1.25	5	3.856	230	215	260	0.009	17.5
0.75	0.55	1425	80M	78.1	78.0	75.1	64.1	72.5	62.0	47.5	30.0	1.40	8	3.680	290	260	305	0.010	17.5
		900	80M	68.5	68.8	64.9	50.2	67.0	57.0	44.0	29.0	1.73	7	5.827	225	220	250	0.012	19.5
1	0.75	2850	80M	77.4	78.0	76.3	64.3	85.5	78.5	66.0	44.5	1.64	9	2.509	215	180	280	0.005	17.0
		1415	80M	79.6	79.5	76.9	66.3	73.5	63.5	49.5	31.0	1.85	11	5.054	300	330	325	0.013	20.5
		935	90S	75.9	76.4	73.9	63.8	69.5	60.0	46.5	29.5	2.05	10	7.649	210	185	260	0.019	25.5
		695	100L	71.8	71.0	68.0	54.0	65.0	56.0	43.5	28.0	2.32	10	10.29	210	175	235	0.046	37.5
1.5	1.1	2875	80M	79.6	80.0	78.3	68.5	85.5	79.0	67.0	45.0	2.33	17	3.648	255	200	305	0.007	19.5
		1445	90S	81.4	81.4	78.9	69.8	76.0	67.0	53.0	33.5	2.57	19	7.259	270	205	325	0.017	25.0
		930	90L	78.1	78.8	76.9	68.2	71.5	62.0	48.5	30.5	2.84	14	11.28	215	190	260	0.026	30.0
		690	100L	74.7	75.0	73.0	61.5	67.5	58.5	45.5	28.0	3.15	14	15.20	210	175	230	0.059	44.5
2	1.5	2880	90S	81.3	81.8	80.3	73.5	86.5	80.5	69.0	48.0	3.08	24	4.966	260	245	325	0.011	24.5
		1435	90L	82.8	83.7	82.6	75.7	81.0	73.0	59.5	38.0	3.23	23	9.967	250	180	300	0.022	28.0
		950	100L	79.8	80.5	78.8	68.5	70.5	61.5	48.5	30.0	3.85	19	15.06	170	140	240	0.048	39.0
		700	112M	76.8	77.0	75.5	63.0	66.0	57.0	45.0	28.0	4.27	18	20.43	200	150	225	0.071	49.5
3	2.2	2875	90L	83.2	84.3	83.4	77.9	87.5	82.0	70.5	48.5	4.36	35	7.297	285	240	335	0.014	28.0
		1450	100L	84.3	85.0	84.1	76.1	81.5	74.0	61.0	39.0	4.62	33	14.47	210	170	300	0.041	37.0
		950	112M	81.8	82.4	81.1	72.6	75.0	66.5	53.0	33.5	5.18	34	22.08	280	255	300	0.071	49.0
		710	132S	79.4	82.0	79.5	69.0	64.5	55.0	42.0	25.0	6.20	31	29.55	240	235	300	0.138	65.5
4	3	2895	100L	84.6	85.9	85.7	80.4	88.0	83.0	73.0	50.0	5.82	49	9.88	245	225	310	0.022	37.6
		1445	100L	85.5	85.9	84.8	77.3	82.0	75.0	62.5	40.0	6.18	44	19.80	210	170	300	0.050	40.0
		960	132S	83.3	84.1	83.2	76.8	78.0	71.0	58.0	37.0	6.66	37	29.80	190	165	300	0.103	61.0
		700	132M	81.3	83.0	81.5	72.0	69.0	59.5	46.0	28.0	7.72	37	40.87	215	210	270	0.162	71.0
5.5	4	2880	112M	85.8	86.9	86.6	81.4	91.0	88.0	81.0	61.5	7.39	63	13.24	235	240	335	0.042	49.0
		1450	112M	86.6	87.6	87.5	83.2	85.0	80.5	71.0	48.0	7.84	58	26.30	220	200	300	0.083	54.0
		960	132M	84.6	85.6	85.1	79.3	79.0	72.5	60.0	38.5	8.64	53	39.73	210	180	300	0.131	69.0
		715	160M	83.0	84.0	82.0	73.5	71.5	63.0	51.0	31.0	9.73	55	53.34	185	160	270	0.343	110
7.5	5.5	2925	132S	87.0	87.2	86.2	81.0	86.0	82.5	74.5	55.5	10.6	82	17.93	240	180	300	0.063	68.0
		1455	132S	87.7	88.7	88.6	84.5	85.5	80.5	70.0	47.0	10.6	81	36.04	255	210	305	0.123	72.0
		960	132M	86.0	86.9	86.5	81.2	79.5	72.5	60.5	38.5	11.6	78	54.63	230	195	300	0.188	81.0
		715	160M	84.5	84.0	82.5	74.0	71.0	63.0	51.0	31.0	13.2	70	73.35	185	160	265	0.343	111

ELECTRICAL CHARACTERISTICS - AESV2E / AESU2E

IE2

TEFC, CLASS F, 40°C AMBIENT TEMP., IEC DESIGN N CONTINUOUS DUTY, S.F. 1.0

400V/50HZ

OUTPUT		FULL LOAD rpm	FRAME NO.	EFFICIENCY				POWER FACTOR				CURRENT		TORQUE				ROTOR GD2 kg-m2	APPROX. WEIGHT kg
HP	kW			FULL LOAD (%)	3/4 LOAD (%)	2/4 LOAD (%)	1/4 LOAD (%)	FULL LOAD (%)	3/4 LOAD (%)	2/4 LOAD (%)	1/4 LOAD (%)	FULL LOAD (A)	LOCKED ROTOR (A)	FULL LOAD N·m	LOCKED ROTOR %FLT	PULL UP %FLT	BREAK DOWN %FLT		
10	7.5	2920	132S	88.1	88.3	87.8	83.3	82.5	77.5	68.0	46.5	14.9	98	24.49	250	230	275	0.075	72.5
		1460	132M	88.7	89.6	89.5	85.3	84.0	78.5	67.0	44.5	14.5	112	48.98	275	200	305	0.142	79.0
		960	160M	87.2	88.2	87.7	82.4	82.0	77.0	66.5	44.5	15.1	105	74.50	210	195	260	0.363	110
		720	160L	86.0	86.0	84.0	76.0	70.0	61.0	49.0	30.0	18.0	105	99.33	210	180	300	0.586	146
15	11	2950	160M	89.4	89.3	88.1	82.5	90.5	87.5	80.5	61.5	19.6	172	35.56	230	180	305	0.154	110
		1465	160M	89.8	90.6	90.7	87.1	86.5	83.0	74.5	52.5	20.4	160	71.60	220	180	300	0.296	121
		965	160L	88.7	89.2	88.6	83.2	81.5	76.0	65.0	42.5	22.0	170	108.7	245	205	300	0.558	138
		720	180L	87.7	87.5	87.0	80.0	70.0	62.0	56.0	34.0	25.9	140	145.7	210	160	230	1.019	182
20	15	2930	160M	90.3	91.0	91.2	88.1	93.5	92.5	89.0	74.0	25.6	225	48.82	245	165	280	0.192	120
		1470	160L	90.6	91.3	91.2	88.1	86.5	82.5	73.5	51.5	27.6	220	97.30	220	185	300	0.427	138
		975	180L	89.7	90.4	90.2	86.7	82.5	77.5	67.5	46.0	29.3	220	146.7	210	195	300	1.337	205
		720	200L	89.0	90.0	91.0	87.0	77.0	71.5	64.5	41.0	31.6	165	198.7	185	140	205	1.749	275
25	18.5	2925	160L	90.9	91.5	91.7	88.6	93.0	91.5	88.0	73.0	31.6	290	60.31	260	185	310	0.237	137
		1475	180M	91.2	91.7	91.6	88.7	85.5	83.0	76.5	57.0	34.2	230	119.6	200	185	300	0.654	180
		975	200L	90.4	91.0	90.9	87.7	79.5	75.0	65.5	43.5	37.2	260	180.0	215	195	300	1.604	263
		735	225S	91.5	92.0	91.0	86.0	72.0	65.5	58.0	35.5	40.5	220	240.0	210	185	235	2.675	345
30	22	2930	180M	91.3	91.2	90.5	85.8	91.5	90.0	85.5	71.0	38.0	295	71.60	215	185	300	0.283	178
		1470	180L	91.6	92.4	92.2	89.3	85.5	83.5	77.5	58.0	40.5	270	142.7	195	155	250	0.770	199
		980	200L	90.9	91.4	91.8	88.0	81.0	77.5	68.5	44.0	43.1	300	214.1	210	180	255	1.912	283
		735	225M	92.0	92.0	92.0	88.0	74.5	69.0	63.0	39.5	46.3	240	285.4	210	170	215	3.023	367
40	30	2945	200L	92.0	92.2	91.3	86.3	90.5	89.5	86.0	71.5	52.0	440	97.13	210	150	300	0.521	276
		1470	200L	92.3	92.9	92.9	90.6	87.5	84.5	77.0	56.0	53.6	420	194.6	230	180	300	1.217	266
		980	225M	91.7	92.4	92.2	88.9	86.0	83.0	76.0	55.5	54.9	365	291.9	210	190	285	2.442	343
		735	250M	92.0	92.0	92.0	88.0	74.5	68.0	58.0	36.0	63.2	350	389.2	210	170	245	4.565	475
50	37	2945	200L	92.5	92.9	92.7	89.1	91.5	90.5	87.5	75.0	63.1	585	119.8	175	130	260	0.663	302
		1475	225S	92.7	93.3	93.3	90.3	87.5	86.0	80.0	61.5	65.8	430	239.2	220	175	260	1.649	333
		980	250M	92.2	92.3	91.7	87.2	86.5	83.0	74.0	50.5	67.0	455	360.0	210	185	275	3.373	458
		736	280S	92.2	92.3	91.5	88.2	81.0	77.0	67.5	45.8	71.5	455	480.1	150	130	270	8.400	645
60	45	2965	225M	92.9	92.5	91.3	85.9	91.0	88.5	82.5	64.5	76.8	710	144.7	170	140	300	1.074	333
		1480	225M	93.1	93.3	92.9	89.8	86.0	82.5	75.0	53.5	81.1	580	289.9	210	170	300	1.979	368
		985	280S	92.9	92.9	92.2	89.0	84.0	79.5	69.5	46.5	83.2	555	436.3	205	185	295	6.400	580
		736	280M	92.6	92.6	92.1	89.2	81.0	78.0	68.0	47.0	86.6	540	583.9	150	130	270	9.600	690
75	55	2970	250M	93.2	93.2	92.6	88.6	92.5	91.5	88.0	74.0	92.1	730	176.6	165	150	300	1.343	456
		1485	250M	93.5	93.7	93.3	90.0	87.5	85.0	79.0	59.5	97.0	780	353.2	245	180	300	3.621	492
		985	280M	93.1	93.1	92.6	90.0	83.5	79.5	70.0	47.0	102.1	690	533.2	210	185	275	7.600	660
		738	315S	93.0	93.0	92.3	89.4	79.0	75.0	63.5	42.0	108.1	660	711.7	170	150	250	16.00	900

TEFC, CLASS F, 40°C AMBIENT TEMP., IEC DESIGN N CONTINUOUS DUTY, S.F. 1.0
400V/50HZ

OUTPUT		FULL LOAD rpm	FRAME NO.	EFFICIENCY				POWER FACTOR				CURRENT		TORQUE				ROTOR GD2 kg-m2	APPROX. WEIGHT kg
HP	KW			FULL LOAD (%)	3/4 LOAD (%)	2/4 LOAD (%)	1/4 LOAD (%)	FULL LOAD (%)	3/4 LOAD (%)	2/4 LOAD (%)	1/4 LOAD (%)	FULL LOAD (A)	LOCKED ROTOR (A)	FULL LOAD N•m	LOCKED ROTOR %FLT	PULL UP %FLT	BREAK DOWN %FLT		
100	75	2970	280S	93.8	93.8	93.2	89.0	88.5	85.5	77.0	53.5	130.4	990	241.2	210	190	315	2.000	565
		1482	280S	94.0	94.0	93.5	91.8	87.5	86.0	78.5	57.5	131.6	890	483.3	200	180	280	5.200	620
		985	315S	93.7	93.7	93.1	88.8	83.5	80.5	70.0	47.5	138.4	850	727.2	200	180	260	10.80	870
		738	315M	93.7	93.8	93.5	90.4	79.0	75.0	64.5	42.0	146.2	885	970.5	170	150	250	21.20	1,060
125	90	2970	280M	94.1	94.1	93.6	90.4	89.0	86.5	80.0	55.0	155.1	1260	289.4	225	205	310	2.400	615
		1482	280M	94.2	94.2	93.6	91.8	87.5	86.0	79.0	58.0	157.6	1090	580.0	200	180	280	6.400	690
		987	315M	94.0	94.0	93.5	91.0	84.0	80.5	70.5	48.0	164.5	1070	870.8	200	180	260	12.80	970
		740	315L	94.0	94.0	93.6	90.4	80.0	76.0	65.5	44.0	172.7	1050	1161	170	150	250	24.40	1,180
150	110	2975	315S	94.3	94.3	93.7	90.5	88.0	84.0	76.5	54.0	191.3	1520	353.1	205	185	320	3.600	840
		1482	315S	94.5	94.5	93.8	91.8	88.0	85.5	79.0	58.5	190.9	1400	708.8	205	185	280	8.80	940
		988	315L	94.3	94.3	93.8	91.0	85.0	82.0	73.0	48.0	198.1	1370	1063	205	185	270	16.40	1,070
		740	315L	94.5	94.5	94.1	91.0	80.0	76.0	65.5	44.0	210.0	1300	1420	180	160	260	30.80	1,320
175	132	2975	315M	94.6	94.6	93.8	90.6	89.5	87.0	80.0	58.0	225.0	1800	423.7	220	200	320	4.400	920
		1485	315M	94.7	94.7	94.0	91.8	88.0	86.0	80.0	58.5	228.6	1620	848.9	205	185	290	10.00	1,020
		988	315L	94.6	94.6	94.0	91.2	85.0	82.0	73.0	48.0	236.9	1500	1276	205	185	275	18.40	1,250
		741	315D	94.6	94.6	94.1	91.0	80.2	76.3	65.0	43.0	251.1	1606	1701	140	120	270	34.80	1,700
215	160	2975	315M	94.8	94.8	94.2	91.0	90.5	88.5	84.0	65.0	269.2	2175	513.6	225	205	330	5.600	1,000
		1485	315M	94.9	94.9	94.3	92.0	88.5	87.0	80.5	60.5	275.0	1900	1029	200	180	285	11.60	1,100
		988	315L	94.8	94.8	94.1	91.2	85.0	82.0	74.0	48.5	286.6	1880	1547	210	190	275	23.20	1,340
		741	315D	94.8	94.8	94.2	91.2	80.0	75.5	64.0	41.5	304.5	2035	2062	140	120	275	40.80	1,800
270	200	2975	315L	95.0	95.0	94.2	91.2	91.0	90.5	88.5	74.5	333.9	2620	642.0	230	215	315	7.200	1,200
		1485	315L	95.1	95.1	94.3	92.2	89.0	87.5	82.0	62.0	341.1	2400	1286	200	180	285	14.40	1,280
		988	315D	95.0	95.0	94.2	91.2	85.5	84.0	76.0	55.0	355.4	2540	1933	210	190	280	38.80	1,665
		741	315D	95.0	95.0	94.3	91.5	80.5	76.2	65.8	42.0	377.5	2445	2578	140	120	270	47.20	1,950
335	250	2977	315D	95.0	95.0	94.3	91.5	91.5	91.0	88.5	76.0	415.1	3110	802.0	180	160	300	10.00	1,600
		1485	315D	95.1	95.1	94.4	92.4	90.0	89.0	84.5	69.0	421.6	3150	1608	210	185	280	25.20	1,650
		988	315D	95.0	95.0	94.3	91.5	86.5	85.0	78.5	58.0	439.1	3040	2416	200	180	270	46.40	1,840
420	315	2977	315D	95.0	95.0	94.3	91.7	92.5	92.0	90.0	76.5	517.4	3760	1010	170	150	290	12.00	1,760
		1485	315D	95.1	95.1	94.4	92.5	90.5	89.5	86.0	70.0	528.3	3670	2026	210	190	260	31.20	1,860

Note : 1. The above are typical values based on test according to IEC 60034-2-1.

2. Tolerance according to IEC 60034-1.

3. Efficiency, power factor, speed and torque are the same for other voltages.

Current values vary inversely with voltage.

4. 0.55 kW and below, 315 kW up, and all 8 Pole rating : Efficiency per TECO performance standard.

5. Data subject to change without notice.

ELECTRICAL CHARACTERISTICS - AESV2E / AESU2E

IE2

TEFC, CLASS F, 40°C AMBIENT TEMP., IEC DESIGN N CONTINUOUS DUTY,
S.F. 1.0 50HZ 400V WINDING USED ON 60HZ 460V PLUS B RISE

400V/50HZ

OUTPUT		FULL LOAD rpm	FRAME NO.	EFFICIENCY				POWER FACTOR				CURRENT		TORQUE				ROTOR GD2 kg-m2	APPROX. WEIGHT kg
HP	kW			FULL LOAD (%)	3/4 LOAD (%)	2/4 LOAD (%)	1/4 LOAD (%)	FULL LOAD (%)	3/4 LOAD (%)	2/4 LOAD (%)	1/4 LOAD (%)	FULL LOAD (A)	LOCKED ROTOR (A)	FULL LOAD N•m	LOCKED ROTOR %FLT	PULL UP %FLT	BREAK DOWN %FLT		
0.58	0.43	1115	80M	67.5	66.2	60.4	44.3	62.5	53.0	41.5	28.0	1.28	5	3.683	230	215	260	0.009	17.5
0.84	0.63	1730	80M	81.2	79.8	76.1	63.8	71.5	62.5	48.5	31.0	1.36	10	3.478	315	280	345	0.010	17.5
		1110	80M	72.5	72.5	68.5	54.5	67.5	58.0	43.5	28.5	1.62	7	5.420	230	225	265	0.012	19.5
1.15	0.86	3450	80M	80.7	81.0	79.0	69.4	84.5	79.0	67.0	47.0	1.58	11	2.381	280	215	300	0.005	17.0
		1715	80M	82.5	81.1	77.9	67.8	70.0	60.5	46.5	29.5	1.87	13	4.789	305	280	350	0.013	20.5
		1140	90S	80.0	78.9	75.9	64.5	70.0	60.5	47.5	30.0	1.93	10	7.204	205	185	270	0.019	25.5
		850	100L	75.0	75.5	73.0	61.0	64.5	55.0	43.0	27.0	2.23	11	9.662	200	165	250	0.046	37.5
1.7	1.27	3465	80M	82.5	82.5	81.0	72.2	85.5	80.0	69.0	48.5	2.26	17	3.500	315	240	325	0.007	19.5
		1740	90S	84.0	83.6	81.0	70.9	77.5	69.5	56.0	36.0	2.45	20	6.970	275	220	335	0.017	25.0
		1140	90L	81.0	80.5	78.0	67.5	71.0	62.0	49.0	31.0	2.77	15	10.64	205	185	270	0.026	30.0
		845	100L	77.5	79.0	78.0	67.0	67.0	58.5	46.0	28.0	3.07	14	14.35	200	160	235	0.059	44.5
2.32	1.73	3475	90S	84.0	84.5	83.2	76.5	85.0	79.0	67.5	48.0	3.04	25	4.754	210	205	300	0.011	24.5
		1730	90L	84.0	84.7	83.7	76.9	80.0	73.5	61.0	40.0	3.23	24	9.550	250	160	300	0.022	28.0
		1155	100L	82.5	82.6	80.5	70.5	71.0	63.0	50.5	31.5	3.71	20	14.30	180	160	260	0.048	39.0
		850	112M	76.5	77.0	76.0	64.5	64.5	56.0	43.5	27.0	4.40	21	19.44	190	110	240	0.071	49.5
3.39	2.53	3480	90L	85.5	85.9	84.6	78.1	86.0	80.5	69.5	50.0	4.32	39	6.943	320	220	330	0.014	28.0
		1755	100L	87.5	87.3	86.5	79.2	81.0	74.0	61.5	33.0	4.48	34	13.77	205	155	300	0.041	37.0
		1155	112M	85.0	85.5	84.3	76.2	73.0	65.0	52.0	31.5	5.12	35	20.92	295	285	310	0.071	49.0
		855	132S	82.5	82.0	79.0	68.5	64.5	55.0	42.5	26.0	5.97	32	28.26	230	215	290	0.138	65.5
4.62	3.45	3490	100L	87.5	87.1	86.6	80.4	88.0	84.0	75.0	55.0	5.62	47	9.441	220	215	355	0.022	37.6
		1745	100L	87.5	87.9	87.4	80.9	82.0	76.0	64.0	41.0	6.04	46	18.88	200	140	300	0.050	40.0
		1165	132S	87.5	87.2	85.9	78.7	78.0	71.5	59.5	38.0	6.34	38	28.28	180	155	285	0.103	61.0
		850	132M	83.0	83.5	82.0	73.0	69.0	60.0	47.0	29.0	7.56	37	38.76	205	190	250	0.162	71.0
6.17	4.6	3485	112M	87.5	87.2	86.1	79.0	91.0	88.0	81.5	63.5	7.25	59	12.61	310	240	345	0.042	49.0
		1750	112M	89.5	89.5	88.8	83.4	84.0	79.0	69.0	47.5	7.68	60	25.10	215	195	325	0.083	54.0
		1165	132M	89.0	88.7	87.9	81.7	78.0	72.0	60.0	38.0	8.32	52	37.71	200	155	290	0.131	69.0
		865	160M	86.0	86.5	86.5	80.0	73.0	64.0	51.0	31.0	9.20	53	50.79	200	170	280	0.343	110
8.49	6.33	3525	132S	88.5	88.2	86.7	79.7	86.5	83.5	76.0	57.5	10.4	79	17.15	210	190	300	0.063	68.0
		1760	132S	89.5	89.8	88.9	83.2	83.5	78.0	66.5	44.0	10.6	85	34.35	270	205	300	0.123	72.0
		1165	132M	89.5	89.4	88.5	82.5	79.5	73.5	62.0	39.5	11.2	72	51.89	200	165	300	0.188	81.0
		865	160M	86.5	87.0	87.0	80.5	71.0	62.5	49.0	29.0	12.9	71	69.89	200	175	270	0.343	111

ELECTRICAL CHARACTERISTICS - AESV2E / AESU2E

**TEFC, CLASS F, 40°C AMBIENT TEMP., IEC DESIGN N CONTINUOUS DUTY,
S.F. 1.0 50HZ 400V WINDING USED ON 60HZ 460V PLUS B RISE**

400V/50HZ

OUTPUT		FULL LOAD rpm	FRAME NO.	EFFICIENCY				POWER FACTOR				CURRENT		TORQUE				ROTOR GD2 kg-m2	APPROX. WEIGHT kg
HP	KW			FULL LOAD (%)	3/4 LOAD (%)	2/4 LOAD (%)	1/4 LOAD (%)	FULL LOAD (%)	3/4 LOAD (%)	2/4 LOAD (%)	1/4 LOAD (%)	FULL LOAD (A)	LOCKED ROTOR (A)	FULL LOAD N·m	LOCKED ROTOR %FLT	PULL UP %FLT	BREAK DOWN %FLT		
11.5	8.6	3525	132S	89.5	89.7	88.9	83.6	86.5	84.0	76.5	57.5	13.9	95	23.30	215	180	265	0.075	72.5
		1755	132M	89.5	89.9	89.4	84.6	85.0	80.5	70.5	47.5	14.2	114	46.80	265	200	300	0.142	79.0
		1160	160M	89.5	89.7	89.0	83.4	82.5	78.5	69.5	46.5	14.6	83	70.80	215	180	255	0.363	110
		870	160L	88.0	88.5	87.0	80.0	74.0	66.0	53.0	32.0	16.6	105	94.40	210	205	350	0.586	146
17	12.7	3545	160M	90.2	90.0	88.5	81.9	91.0	89.0	83.5	66.0	19.4	161	34.21	240	190	300	0.154	110
		1765	160M	91.0	91.5	91.5	87.4	86.5	83.5	76.0	55.5	20.3	142	68.72	240	205	300	0.296	121
		1165	160L	90.2	90.5	89.8	84.7	82.0	76.5	66.0	44.0	21.6	151	104.1	260	215	275	0.558	138
		870	180L	88.0	88.5	88.5	83.0	73.0	67.0	54.5	33.0	24.8	142	139.4	200	170	240	1.019	182
23	17.3	3535	160M	91.0	91.5	90.9	85.8	92.5	91.5	88.0	76.0	25.8	197	46.74	265	255	360	0.192	120
		1765	160L	92.4	92.5	92.5	89.6	87.0	83.5	75.5	54.0	27.0	205	93.61	255	200	300	0.427	138
		1175	180L	91.7	91.5	91.4	87.1	83.5	80.5	72.5	50.5	28.4	192	140.6	245	215	295	1.337	205
		870	200L	89.5	91.0	91.0	87.5	81.0	78.0	69.0	47.5	30.0	164	189.9	190	180	265	1.749	275
28.5	21.3	3535	160L	91.0	91.5	91.0	86.7	92.5	91.5	88.0	76.0	31.8	253	57.54	275	270	370	0.237	137
		1765	180M	92.4	92.6	92.3	89.0	85.5	83.5	77.5	59.5	33.8	215	115.2	200	195	300	0.654	180
		1175	200L	91.7	92.4	92.2	88.7	80.5	77.5	69.0	47.0	36.2	219	173.1	240	220	300	1.604	263
		880	225S	92.0	92.5	92.0	87.5	76.5	72.0	61.5	40.0	38.0	196	231.2	200	140	235	2.675	345
34	25.3	3540	180M	91.0	90.9	89.8	84.2	91.5	90.5	87.0	74.0	38.1	284	68.25	210	190	325	0.283	178
		1770	180L	92.4	93.0	92.7	89.1	85.0	83.5	78.0	58.5	40.4	238	136.5	200	195	300	0.770	199
		1175	200L	91.7	92.4	92.3	89.1	81.5	78.0	70.0	48.0	42.5	260	205.6	240	195	265	1.912	283
		880	225M	92.0	92.5	92.0	88.0	78.0	75.0	66.0	45.0	44.3	208	274.6	180	150	250	3.023	367
46	34.5	3545	200L	92.4	91.6	90.2	84.2	90.5	90.0	86.5	74.5	51.8	404	92.94	220	150	300	0.521	276
		1770	200L	93.0	93.5	93.3	90.4	88.0	86.0	80.0	59.0	52.9	416	186.1	270	240	310	1.217	266
		1180	225M	93.0	93.5	93.2	89.6	86.0	83.5	77.0	56.0	54.1	303	279.2	200	160	245	2.442	343
		885	250M	93.0	93.0	93.0	89.0	79.0	74.5	64.5	41.0	58.9	295	372.3	180	130	225	4.565	475
57	42.6	3550	200L	93.0	92.9	92.0	87.2	91.0	90.0	87.0	75.5	63.2	420	114.6	210	160	265	0.663	302
		1775	225S	93.6	93.5	92.7	88.4	86.5	84.5	78.5	59.5	66.0	410	229.2	205	190	330	1.649	333
		1185	250M	93.6	93.2	92.3	87.7	86.5	84.0	76.5	53.0	66.0	375	343.3	190	145	260	3.373	458
		885	280S	92.8	92.8	92.0	88.0	81.5	78.5	70.0	48.0	70.7	455	459.7	125	105	240	8.400	645
69.5	52	3565	225M	93.0	92.8	91.6	86.1	93.0	92.0	88.0	63.5	75.5	643	139.3	180	180	330	1.074	333
		1780	225M	94.1	93.9	93.2	89.2	86.5	84.0	77.0	56.0	80.2	552	279.0	240	205	320	1.979	368
		1186	280S	93.8	93.6	92.7	89.5	85.0	82.0	73.0	50.0	81.9	550	418.7	185	170	280	6.400	580
		885	280M	93.0	92.8	92.2	88.1	82.0	79.0	71.0	49.0	85.6	550	561.1	120	100	230	9.600	690
84.5	63	3550	250M	93.0	93.1	92.2	87.7	92.0	91.5	88.0	74.5	92.4	650	169.5	160	160	300	1.343	456
		1780	250M	94.1	94.1	93.4	89.8	87.5	85.5	80.0	61.0	96.0	687	338.0	240	180	270	3.621	492
		1186	280M	93.8	93.8	93.5	90.5	86.0	83.0	75.0	52.0	98.0	680	507.3	190	165	280	7.600	660
		887	315S	93.1	93.0	92.5	89.0	81.5	79.0	71.0	50.0	104	518	678.3	130	110	220	16.00	900

ELECTRICAL CHARACTERISTICS - AESV2E / AESU2E

IE2

TEFC, CLASS F, 40°C AMBIENT TEMP., IEC DESIGN N CONTINUOUS DUTY,
S.F. 1.0 50HZ 400V WINDING USED ON 60HZ 460V PLUS B RISE

400V/50HZ

OUTPUT		FULL LOAD rpm	FRAME NO.	EFFICIENCY				POWER FACTOR				CURRENT		TORQUE				ROTOR GD2 kg-m2	APPROX. WEIGHT kg
HP	kW			FULL LOAD (%)	3/4 LOAD (%)	2/4 LOAD (%)	1/4 LOAD (%)	FULL LOAD (%)	3/4 LOAD (%)	2/4 LOAD (%)	1/4 LOAD (%)	FULL LOAD (A)	LOCKED ROTOR (A)	FULL LOAD N·m	LOCKED ROTOR %FLT	PULL UP %FLT	BREAK DOWN %FLT		
115	86	3570	280S	93.7*	93.6	93.5	90.5	88.5	86.0	79.0	58.0	130	980	230.1	190	165	300	2.000	565
		1780	280S	94.5	94.5	93.8	91.5	88.0	87.0	81.0	61.5	130	880	461.4	200	175	280	5.200	620
		1186	315S	94.2	94.3	94.0	91.0	84.0	81.0	72.5	51.0	136	825	692.5	200	175	240	10.80	870
		885	315M	93.7	93.7	92.8	89.5	81.5	79.5	70.0	51.0	141	750	928.0	130	110	220	21.20	1,060
140	104	3570	280M	94.6	94.5	94.0	91.5	89.0	88.0	82.0	62.0	155	1250	278.2	200	175	300	2.400	615
		1782	280M	94.6*	94.6	93.8	91.5	88.5	88.5	83.0	66.0	156	1080	557.4	200	175	260	6.400	690
		1185	315M	94.3*	94.4	94.0	91.8	85.0	82.0	74.5	52.5	163	1030	838.1	190	170	240	12.80	970
		886	315L	94.2	94.1	93.2	90.1	81.5	79.5	72.0	51.5	170	920	1121	135	115	220	24.40	1,180
170	127	3575	315S	94.6	94.3	93.6	90.5	89.0	87.0	80.5	60.0	189	1490	339.3	185	165	290	3.600	840
		1785	315S	95.0	94.8	94.0	91.5	88.5	87.0	81.0	62.5	190	1380	679.5	200	175	280	8.800	940
		1185	315M	95.0	94.8	94.2	91.5	86.0	84.0	77.0	56.0	195	1280	1024	200	175	240	16.40	1,070
		888	315L	94.5	94.4	93.6	90.8	81.5	79.5	72.0	51.5	207	1250	1366	140	120	220	30.80	1,320
204	152	3575	315M	95.0	94.8	94.3	91.5	90.5	88.5	83.5	65.5	222	1780	406.0	200	175	300	4.400	920
		1785	315M	95.0	95.0	94.3	92.2	89.0	87.5	82.0	64.0	226	1590	813.2	200	175	280	10.00	1,020
		1185	315L	95.0	95.0	94.3	92.4	86.0	84.0	76.0	55.0	234	1590	1225	210	185	240	18.40	1,250
		888	315D	94.8	94.8	94.0	91.0	81.5	79.0	70.0	48.0	247	1540	1635	95	75	240	34.80	1,700
247	184	3575	315M	95.2*	95.1	94.5	92.0	91.5	90.5	87.0	72.0	265	2160	491.5	205	180	280	5.600	1,000
		1785	315M	95.0*	95.0	94.6	92.5	89.5	88.0	83.0	65.0	272	1940	984.4	200	175	280	11.60	1,100
		1186	315L	95.0	95.0	94.5	92.5	86.0	84.0	77.5	57.0	283	1860	1482	220	195	255	23.20	1,340
		888	315D	95.0	94.9	94.5	91.0	81.5	78.5	70.0	46.0	298	1932	1979	95	75	250	40.80	1,800
308	230	3572	315L	95.4	95.0	93.6	89.8	92.0	91.5	89.0	77.0	329	2620	614.9	225	200	275	7.200	1,200
		1785	315L	95.4	95.3	94.2	91.5	89.5	88.5	84.0	68.0	338	2410	1231	210	185	275	14.40	1,280
		1185	315D	95.1	95.0	94.2	91.5	86.5	85.0	80.0	64.0	351	2450	1854	200	175	250	38.80	1,800
		888	315D	95.1	95.1	94.5	91.5	82.0	79.0	72.0	48.0	370	2310	2474	100	80	240	47.20	1,950
385	288	3573	315D	95.5	95.2	94.0	90.5	92.0	91.5	89.5	77.0	411	2930	769.8	140	120	260	10.00	1,700
		1785	315D	95.5	95.3	94.4	91.8	90.0	89.5	85.0	70.0	421	3085	1541	200	175	250	25.20	1,800
		1185	315D	95.2	95.2	94.4	92.0	87.0	85.5	80.5	64.0	436	3070	2321	205	180	250	46.40	1,900
485	362	3575	315D	95.6	95.4	94.2	90.5	92.0	91.5	89.5	77.0	517	4125	967.0	160	140	300	12.00	1,820
		1785	315D	95.5	95.3	94.4	91.8	90.0	89.5	85.5	71.0	529	3800	1937	200	175	250	31.20	1,920

Note : 1. The above are typical values based on test according to IEC 60034-2-1:2007.

2. Tolerance according to IEC 60034-1.

3. Efficiency, power factor, speed and torque are the same for other voltages.
Current values vary inversely with voltage.

4. (*) Efficiency, per TECO performance standard (Not IE2).

5. 0.63 kW and below, and all 8 Pole rating : Efficiency per TECO performance standard.

6. Data subject to change without notice.

DIMENSION

Horizontal Foot Mounting B3 (IM 1001)

TYPE: AESV2E, AESV3E

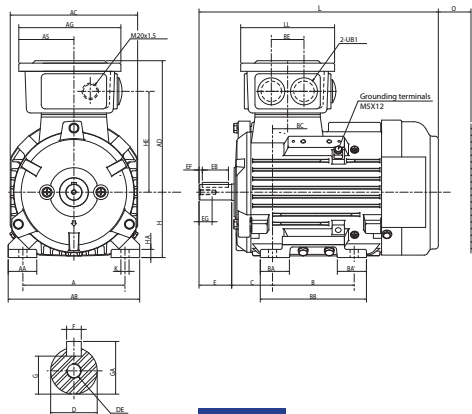


FIG. 1

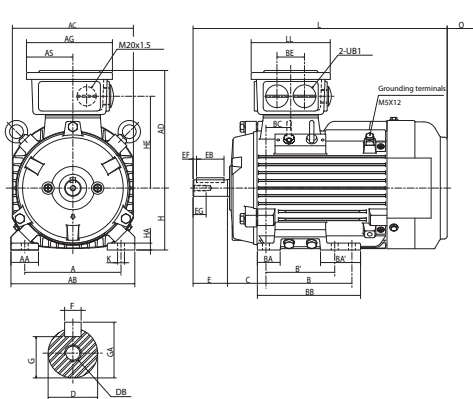


FIG. 2

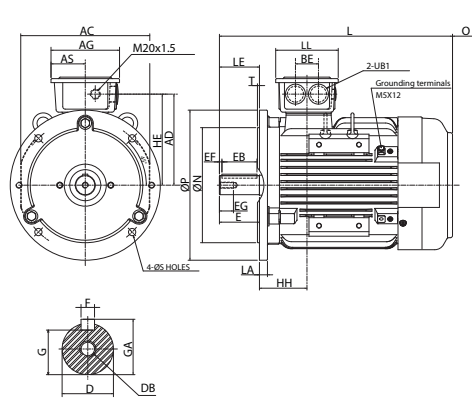


FIG. 3

Dimension in mm

Output (kW)				FRAME SIZE	FIG. NO.	A	AA	AB	AC	AD	AG	AS	B	B'	BA	BA'	BB	BC	BE	C	H
2P	4P	6P	8P																		
0.75	0.55	0.37	—	80M	1	125	35	161	156	161	125	67.5	100	—	35.5	35.5	130	18.5	40	50	80
1.1	0.75	0.55	—																		
1.5	1.1	0.75	—	90S	2	140	40	180	176	171	125	67.5	100	—	33	33	125	36	40	56	90
2.2	1.5	1.1	—	90L		140	40	180	176	171	125	67.5	125	100	33	58	150	36	40	56	90
3	2.2	1.5	0.75	100L		160	40	200	196	191	147	78.5	140	—	43.5	43.5	176	21	50	63	100
—	3	—	1.1																		
4	4	2.2	1.5	112M		190	50	235	218	198.5	147	78.5	140	—	45.5	45.5	176	28	50	70	112
5.5	5.5	3	2.2	132S	3	216	63.5	259	258	216	147	78.5	140	—	59	59	184	6	50	89	132
7.5	—	—	—																		
—	7.5	4	3	132M		216	63.5	259	258	216	147	78.5	178	140	59	97	222	6	50	89	132
—	—	5.5	—																		
11 15	11	7.5	4 5.5	160M	4	254	71	300	317	271	193	91.5	210	—	46	46	256	38	89	108	160
18.5	15	11	7.5	160L		254	71	300	317	271	193	91.5	254	210	46	90	300	38	89	108	160
22	18.5	—	—	180M	5	279	72	330	354	297	193	91.5	241	—	57	57	292	34	89	121	180
—	22	15	11	180L		279	72	330	354	297	193	91.5	279	241	57	95	330	34	89	121	180
30 37	30	18.5 22	15	200L	6	318	88	378	398	330	231	110.5	305	—	70	70	365	53	106	133	200
—	37	—	18.5	225SC		356	94	416	449	356	231	110.5	286	—	70	70	350	30.5	106	149	225
45	—	—	—	225MA		356	94	416	449	356	231	110.5	311	286	70	95	375	30.5	106	149	225
—	45	30	22	225MC		356	94	416	449	356	231	110.5	311	286	70	95	375	30.5	106	149	225
55	—	—	—	250MA		406	112	480	498	398	255	122.5	349	—	84	84	425	45.5	119	168	250
—	55	37	30	250MC		406	112	480	498	398	255	122.5	349	—	84	84	425	45.5	119	168	250

Note : 1. Tolerance of shaft end diameter D: Under $\psi 19 \sim \psi 28 : j6$.
 2. Tolerance of shaft center high H : +0, -0.5.
 3. Tolerance of shaft end diameter D : $\psi 38 : k6$.
 4. Tolerance of shaft center high H : +0, -0.5.

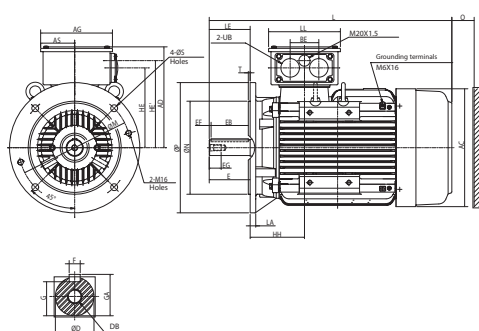


FIG. 4

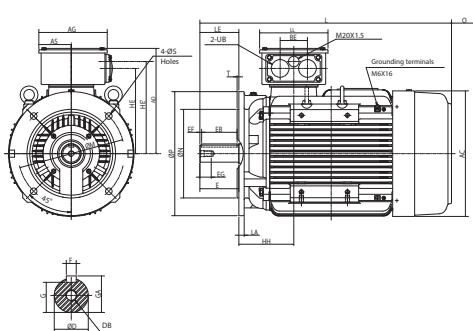


FIG. 5

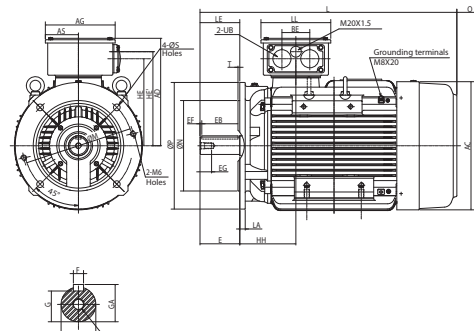


FIG. 6

Dimension in mm

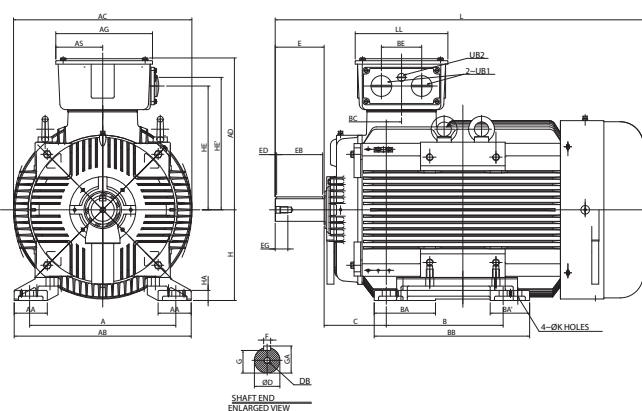
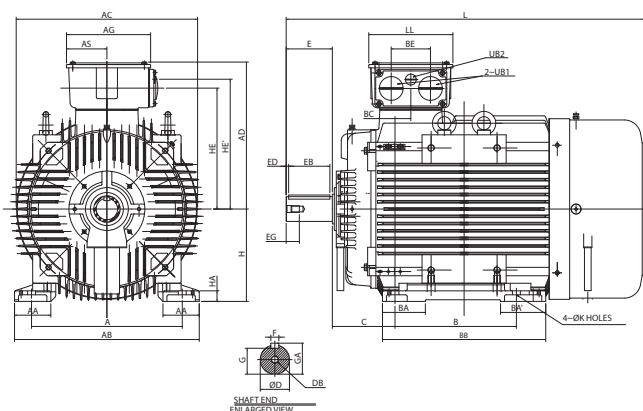
HA	HE	HE'	K	L	LL	O	UB1	SHAFT EXTENSION								BEARING		
								D	E	EB	EF	EG	F	G	GA	DB	DRIVE END	OPPOSITE DRIVE END
10	123.5	—	10	293	115	40	M25X1.5	19	40	32	4	16	6	15.5	21.5	M6	6204ZZC3	6204ZZC3
10	133.5	—	10	344.5	115	40	M25X1.5	24	50	40	5	19	8	20	27	M8	6205ZZC3	6205ZZC3
10	133.5	—	10	369.5	115	40	M25X1.5	24	50	40	5	19	8	20	27	M8	6205ZZC3	6205ZZC3
12	157	—	12	392	125	50	M25X1.5	28	60	50	5	22	8	24	31	M10	6206ZZC3	6206ZZC3
13	164.5	—	12	412.5	125	50	M32X1.5	28	60	50	5	22	8	24	31	M10	6306ZZC3	6306ZZC3
16	182	—	12	466	125	50	M32X1.5	38	80	70	5	28	10	33	41	M12	6308ZZC3	6306ZZC3
16	182	—	12	504	125	50	M32X1.5	38	80	70	5	28	10	33	41	M12	6308ZZC3	6306ZZC3
215	234.5	18	14.5	608	193	60	M40 x 1.5	42	110	100	5	36	12	37	45	M16	6309ZZC3	6307ZZC3
215	234.5	18	14.5	652	193	60	M40 x 1.5	42	110	100	5	36	12	37	45	M16	6309ZZC3	6307ZZC3
241	260.5	20	14.5	672	193	70	M40 x 1.5	48	110	100	5	36	14	42.5	51.5	M16	6311C3	6310C3
241	260.5	20	14.5	710	193	70	M40 x 1.5	48	110	100	5	36	14	42.5	51.5	M16	6311C3	6310C3
24	262	289	18.5	770	231	80	M50 x 1.5	55	110	100	5	42	16	49	59	M20	6312C3	6212C3
28	288	315	18.5	816	231	90	M50 x 1.5	60	140	125	7.5	42	18	53	64	M20	6313C3	6213C3
28	288	315	18.5	811	231	90	M50 x 1.5	55	110	100	5	42	16	49	59	M20	6312C3	6212C3
28	288	315	18.5	841	231	90	M50 x 1.5	60	140	125	7.5	42	18	53	64	M20	6313C3	6213C3
30	322	349	24	921	255	105	M63 x 1.5	60	140	125	7.5	42	18	53	64	M20	6313C3	6313C3
30	322	349	24	921	255	105	M63 x 1.5	65	140	125	7.5	42	18	58	69	M20	6315C3	6313C3

Note : 5. Tolerance of shaft end diameter D : Under $\psi 42 \sim \psi 48$: k6.
6. Tolerance of shaft center high H : +0, -0.5.
7. Tolerance of shaft end diameter D : Under $\psi 55 \sim \psi 65$: m6.
8. Tolerance of shaft center high H : +0, -0.5.

DIMENSION

Horizontal Foot Mounting B3 (IM 1001)

TYPE: AESV2E, AESV3E



Dimension in mm

Output (kW)				FRAME SIZE	FIG. NO.	A	AA	AB	AC	AD	AG	AS	B	B'	BA	BA'	BB	BC	BE	C	H	HA
2P	4P	6P	8P																			
75	—	—	—	280SA	7	457	110	560	550	446	255	122.5	368	—	130	130	445	48	119	190	280	32
—	75	45	37	280SB		457	110	560	550	446	255	122.5	368	—	130	130	445	48	119	190	280	32
—	75	45	37	280SC		457	110	560	550	446	255	122.5	368	—	130	130	445	48	119	190	280	32
90	—	—	—	280MA		457	110	560	550	446	255	122.5	419	—	130	137	495	48	119	190	280	32
—	90	55	45	280MB		457	110	560	550	446	255	122.5	419	—	130	137	495	48	119	190	280	32
—	90	55	45	280MC		457	110	560	550	446	255	122.5	419	—	130	137	495	48	119	190	280	32
110	—	—	—	315SA	8	508	115	615	620	527	336	163	406	—	150	150	508	53	140	216	315	35
—	110	75	55	315SB		508	115	615	620	527	336	163	406	—	150	150	508	53	140	216	315	35
—	110	75	55	315SC		508	115	615	620	527	336	163	406	—	150	150	508	53	140	216	315	35
132 160	—	—	—	315MA		508	115	615	620	527	336	163	457	—	213	137	540	53	140	216	315	35
—	132 160	90 110	75	315MB		508	115	615	620	527	336	163	457	—	213	137	540	53	140	216	315	35
—	132 160	90 110	75	315MC		508	115	615	620	527	336	163	457	—	213	137	540	53	140	216	315	35
200	—	—	—	315LA	9	508	150	650	620	527	336	163	508	—	180	205	730	53	140	216	315	45
—	200	132 160	90 110	315LB		508	150	650	620	527	336	163	508	—	180	205	730	53	140	216	315	45
—	200	132 160	90 110	315LC		508	150	650	620	527	336	163	508	—	180	205	730	53	140	216	315	45
250 315	—	—	—	315DA	10	508	150	650	682	590	412	189	900	—	255	255	1080	68	180	216	315	45
—	250 315	200 250	132 160 200	315DB		508	150	650	682	590	412	189	900	—	255	255	1080	68	180	216	315	45
—	250 315	200 250	132 160 200	315DC		508	150	650	682	590	412	189	900	—	255	255	1080	68	180	216	315	45

- Note :
1. Tolerance of Shaft End Diameter D : m6
 2. Tolerance of Shaft Center Height H : +0, -1
 3. Tolerance of Key Width F : h9
 4. Tolerance of Shaft End Diameter D : m6
 5. Tolerance of Shaft Center Height H : +0, -1
 6. Tolerance of Key Width F : h9

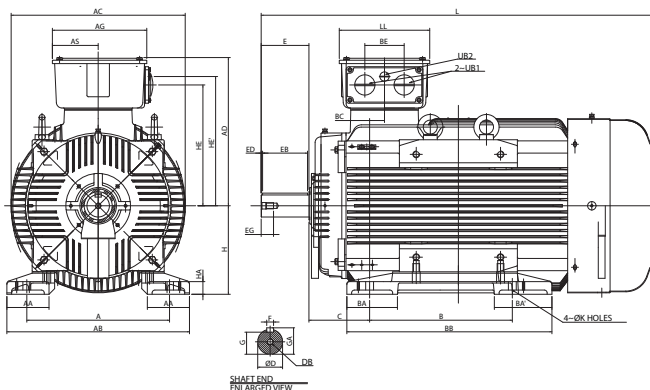


FIG. 9

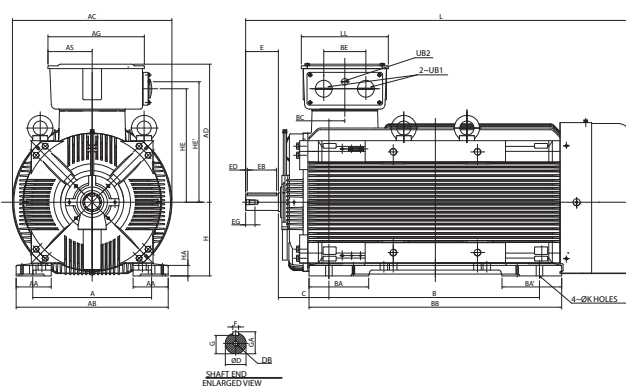


FIG. 10

Dimension in mm

HE	HE'	K	L	LL	O	UB1	UB2	SHAFT EXTENSION								BEARING		
								D	E	EB	ED	EG	F	G	GA	DB	DRIVE END	OPPOSITE DRIVE END
367	394	24	1037	255	140	M63X1.5	M20X1.5	65	140	125	7.5	42	18	58	69	M20	6316C3	6314C3
367	394	24	1037	255	140	M63X1.5	M20X1.5	75	140	125	7.5	42	20	67.5	79.5	M20	6318C3	6316C3
367	394	24	1037	255	140	M63X1.5	M20X1.5	75	140	125	7.5	42	20	67.5	79.5	M20	NU318	6316C3
367	394	24	1087.5	255	140	M63X1.5	M20X1.5	65	140	125	7.5	42	18	58	69	M20	6316C3	6314C3
367	394	24	1087.5	255	140	M63X1.5	M20X1.5	75	140	125	7.5	42	20	67.5	79.5	M20	6318C3	6316C3
367	394	24	1087.5	255	140	M63X1.5	M20X1.5	75	140	125	7.5	42	20	67.5	79.5	M20	NU318	6316C3
430	460	28	1216	322	180	M63X1.5	M20X1.5	65	140	125	7.5	42	18	58	69	M20	6316C3	6314C3
430	460	28	1246	322	180	M63X1.5	M20X1.5	80	170	160	5	42	22	71	85	M20	6320C3	6316C3
430	460	28	1246	322	180	M63X1.5	M20X1.5	80	170	160	5	42	22	71	85	M20	NU320	6316C3
430	460	28	1266	322	180	M63X1.5	M20X1.5	65	140	125	7.5	42	18	58	69	M20	6316C3	6314C3
430	460	28	1296	322	180	M63X1.5	M20X1.5	80	170	160	5	42	22	71	85	M20	6320C3	6316C3
430	460	28	1296	322	180	M63X1.5	M20X1.5	80	170	160	5	42	22	71	85	M20	NU320	6316C3
430	460	28	1366	322	180	M63X1.5	M20X1.5	65	140	125	7.5	42	18	58	69	M20	6316C3	6314C3
430	460	28	1396	322	180	M63X1.5	M20X1.5	80	170	160	5	42	22	71	85	M20	6320C3	6316C3
430	460	28	1396	322	180	M63X1.5	M20X1.5	80	170	160	5	42	22	71	85	M20	NU320	6316C3
485	515	28	1674	372	200	M63X1.5	M20X1.5	65	140	125	7.5	42	18	58	69	M20	6316C3	6316C3
485	515	28	1704	372	200	M63X1.5	M20X1.5	85	170	160	5.0	42	22	76	90	M20	6322C3	6322C3
485	515	28	1704	372	200	M63X1.5	M20X1.5	95	170	160	5.0	50	25	86	100	M24	NU322	6322C3

Note : 7. Tolerance of Shaft End Diameter D : m6

8. Tolerance of Shaft Center Height H : +0, -1

9. Tolerance of Key Width F : h9

11. Tolerance of Shaft End Diameter D : m6

12. Tolerance of Shaft Center Height H : +0, -1

13. Tolerance of Key Width F : h9

DIMENSION

Horizontal Flange Mounting B5 (IM 3001)

Vertical Flange Mounted, Shaft Down V1 (IM 3011)

TYPE: AESU2E, AESU3E

Totally Enclosed Fan-Cooled Type, Squirrel-Cage Rotor.

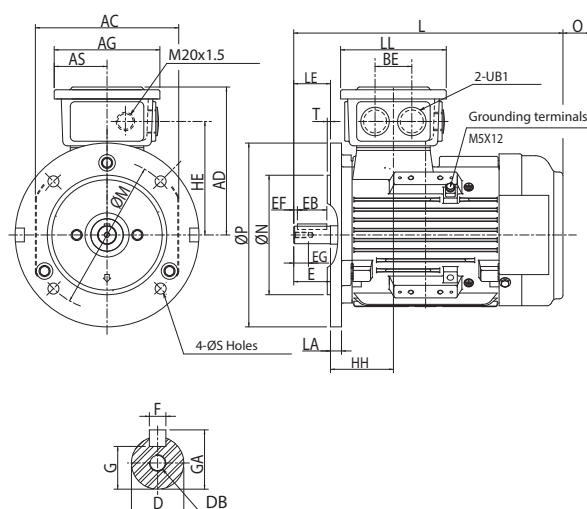


FIG. 1

Dimension in mm

Output (kW)				FRAME SIZE	FIG. NO.	FLANGE DIMENSION							AC	AD	AG	AS	BE	HE
2P	4P	6P	8P			LA	LE	M	N	P	S	T						
0.75	0.55	0.37	—	80M	1	12	40	165	130	200	12	3.5	156	161	125	67.5	40	123.5
1.1	0.75	0.55	—			12	40	165	130	200	12	3.5	156	161	125	67.5	40	123.5
1.5	1.1	0.75	—	90S	2	12	50	165	130	200	12	3.5	176	171	125	67.5	40	133.5
2.2	1.5	1.1	—	90L		12	50	165	130	200	12	3.5	176	171	125	67.5	40	133.5
3	2.2	1.5	0.75	100L		16	60	215	180	250	14.5	4	196	191	147	78.5	50	157
—	3	—	1.1			16	60	215	180	250	14.5	4	196	191	147	78.5	50	157
4	4	2.2	1.5	112M		15	60	215	180	250	14.5	4	218	198.5	147	78.5	50	164.5
5.5	5.5	3	2.2	132S	3	16	80	265	230	300	14.5	4	258	216	147	78.5	50	182
7.5	—	—	—			16	80	265	230	300	14.5	4	258	216	147	78.5	50	182
—	7.5	4	3	132M		16	80	265	230	300	14.5	4	258	216	147	78.5	50	182
—	—	5.5	—			16	80	265	230	300	14.5	4	258	216	147	78.5	50	182

- Note :
1. Tolerance of Shaft End Diameter D : j6
 2. Tolerance of N : j6
 3. Tolerance of shaft end diameter D : $\psi 38:k6$.
 4. Tolerance of N : j6.

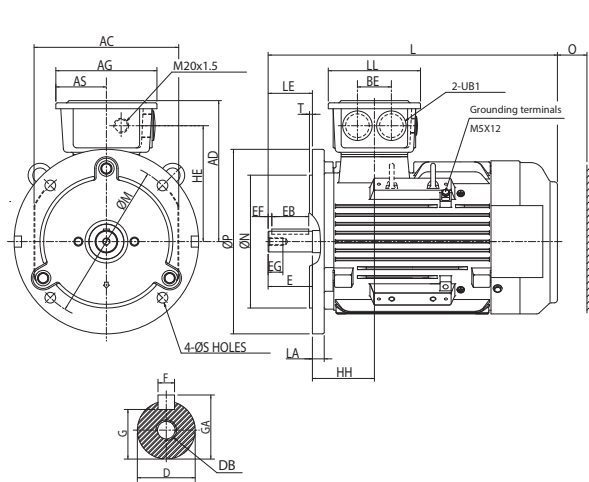


FIG. 2

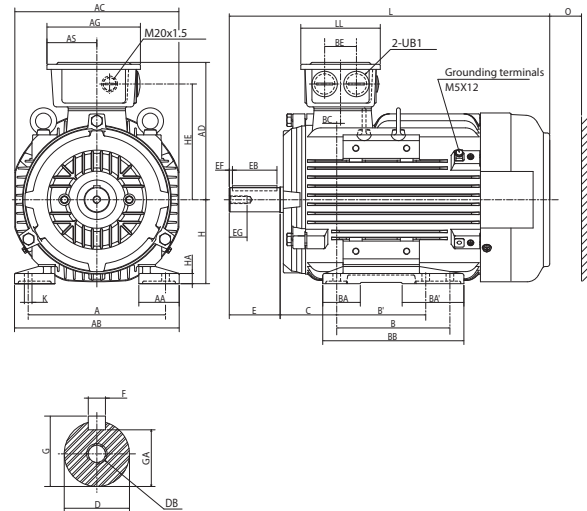


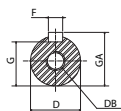
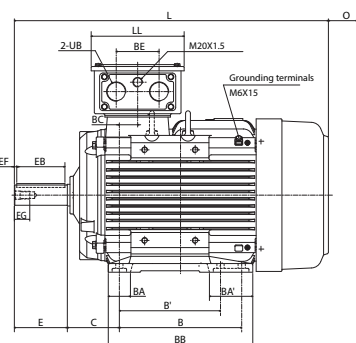
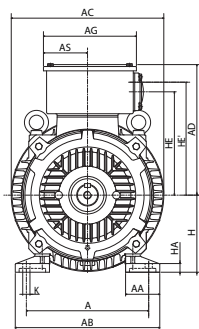
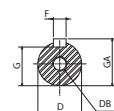
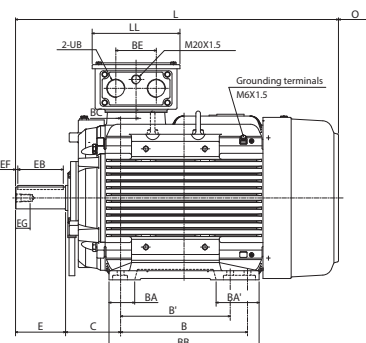
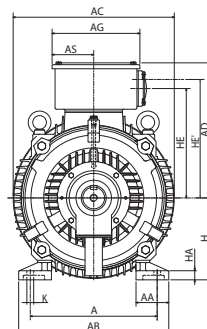
FIG. 3

Dimension in mm

HH	L	LL	O	UB1	SHAFT EXTENSION									BEARING	
					D	E	EB	EF	EG	F	G	GA	DB	DRIVE END	OPPOSITE
68.5	293	115	40	M25X1.5	19	40	32	4	16	6	15.5	21.5	M6	6204ZZC3	6204ZZC3
92	344.5	115	40	M25X1.5	24	50	40	5	19	8	20	27	M8	6205ZZC3	6205ZZC3
92	369.5	115	40	M32X1.5	24	50	40	5	19	8	20	27	M8	6205ZZC3	6205ZZC3
84	392	125	50	M32X1.5	28	60	50	5	22	8	24	31	M10	6206ZZC3	6206ZZC3
98	412.5	125	50	M25X1.5	28	60	50	5	22	8	24	31	M10	6306ZZC3	6306ZZC3
95	466	125	50	M32X1.5	38	80	70	5	28	10	33	41	M12	6308ZZC3	6306ZZC3
95	504	125	50	M32X1.5	38	80	70	5	28	10	33	41	M12	6308ZZC3	6306ZZC3

IE2

IE3

DIMENSION**Horizontal Flange Mounting B5 (IM 3001)****Vertical Flange Mounted, Shaft Down V1 (IM 3011)****TYPE: AESU2E, AESU3E****FIG. 4****FIG. 5***Dimension in mm*

Output (kW)				FRAME SIZE	FIG. NO.	FLANGE DIMENSION							AC	AD	AG	AS	BE	HE	HE'
2P	4P	6P	8P			LA	LE	M	N	P	S	T							
11 15	11	7.5	4 5.5	160M	4	15	110	300	250	350	18.5	5	317	271	193	91.5	89	215	234.5
18.5	15	11	7.5	160L		15	110	300	250	350	18.5	5	317	271	193	91.5	89	215	234.5
22	18.5	—	—	180M	5	15	110	300	250	350	18.5	5	354	297	193	91.5	89	241	260.5
—	22	15	11	180L		15	110	300	250	350	18.5	5	354	297	193	91.5	89	241	260.5
30 37	30	18.5 22	15	200L	6	17	110	350	300	400	18.5	5	398	330	231	110.5	106	262	289
—	37	—	18.5	225SC	7	20	140	400	350	450	18.5	5	449	356	231	110.5	106	288	315
45	—	—	—	225MA		20	110	400	350	450	18.5	5	449	356	231	110.5	106	288	315
—	45	30	22	225MC		20	140	400	350	450	18.5	5	449	356	231	110.5	106	288	315
55	—	—	—	250MA		22	140	500	450	550	18.5	5	498	398	255	122.5	119	322	349
—	55	37	30	250MC		22	140	500	450	550	18.5	5	498	398	255	122.5	119	322	349

Note : 1. Tolerance of shaft end diameter D : Under $\psi 42 \sim \psi 48$: k6.

2. Tolerance of N : j6.

3. Tolerance of shaft end diameter D : Under $\psi 55 \sim \psi 65$: m6.

4. Tolerance of N : j6.

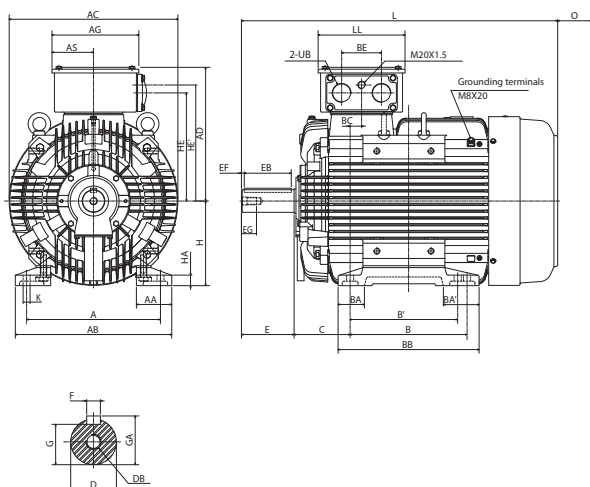


FIG. 6

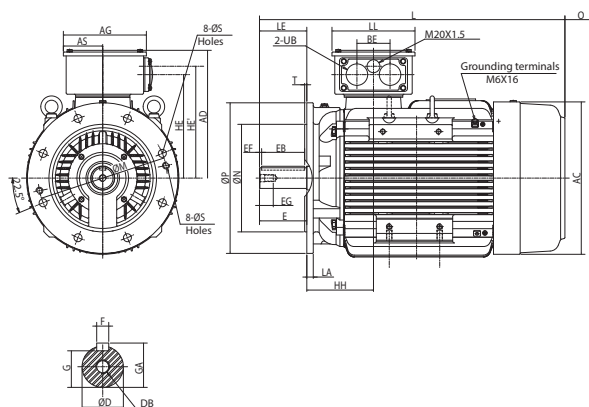


FIG. 7

Dimension in mm

HH	L	LL	O	UB	SHAFT EXTENSION									BEARING	
					D	E	EB	EF	EG	F	G	GA	DB	DRIVE END	OPPOSITE DRIVE END
146	608	193	60	M40 x 1.5	42	110	100	5	32	12	37.0	45.0	M16	6309ZC3	6307ZC3
146	652	193	60	M40 x 1.5	42	110	100	5	32	12	37.0	45.0	M16	6309ZC3	6307ZC3
155	672	193	70	M40 x 1.5	48	110	100	5	32	14	42.5	51.5	M16	6311C3	6310C3
155	710	193	70	M40 x 1.5	48	110	100	5	32	14	42.5	51.5	M16	6311C3	6310C3
193	770	231	80	M50 x 1.5	55	110	100	5.0	42	16	49	59	M20	6312C3	6212C3
179.5	816	231	90	M50 x 1.5	60	140	125	7.5	42	18	53	64	M20	6313C3	6213C3
179.5	811	231	90	M50 x 1.5	55	110	100	5.0	42	16	49	59	M20	6312C3	6212C3
179.5	841	231	90	M50 x 1.5	60	140	125	7.5	42	18	53	64	M20	6313C3	6213C3
213.5	921	255	105	M63 x 1.5	60	140	125	7.5	42	18	53	64	M20	6313C3	6313C3
213.5	921	255	105	M63 x 1.5	65	140	125	7.5	42	18	58	69	M20	6315C3	6313C3

DIMENSION

Horizontal Flange Mounting B5 (IM 3001)

Vertical Flange Mounted, Shaft Down V1 (IM 3011)

TYPE: AESU2E, AESU3E

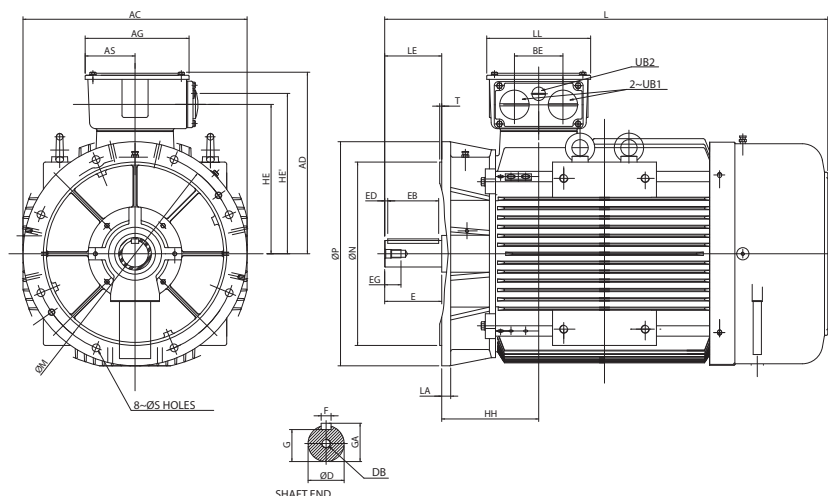


FIG. 8

Dimension in mm

Output (kW)				FRAME SIZE	FIG. NO.	FLANGE DIMENSION							AC	AD	AG	AS	BE	HE	HE'	HH	L	LL	O
2P	4P	6P	8P			LA	LE	M	N	P	S	T											
75	—	—	—	280SA	8	22	140	500	450	550	18.5	5	550	446	255	122.5	119	367	394	238	1037.5	255	140
—	75	45	37	280SB		22	140	500	450	550	18.5	5	550	446	255	122.5	119	367	394	238	1037.5	255	140
—	75	45	37	280SC		22	140	500	450	550	18.5	5	550	446	255	122.5	119	367	394	238	1037.5	255	140
90	—	—	—	280MA		22	140	500	450	550	18.5	5	550	446	255	122.5	119	367	394	238	1087.5	255	140
—	90	55	45	280MB		22	140	500	450	550	18.5	5	550	446	255	122.5	119	367	394	238	1087.5	255	140
—	90	55	45	280MC		22	140	500	450	550	18.5	5	550	446	255	122.5	119	367	394	238	1087.5	255	140
110	—	—	—	315SA	9	25	140	600	550	660	24	6	660	527	336	163	140	430	460	269	1216	322	180
—	110	75	55	315SB		25	170	600	550	660	24	6	660	527	336	163	140	430	460	269	1246	322	180
—	110	75	55	315SC		25	170	600	550	660	24	6	660	527	336	163	140	430	460	269	1246	322	180
132 160	—	—	—	315MA		25	140	600	550	660	24	6	660	527	336	163	140	430	460	269	1266	322	180
—	132 160	90 110	75	315MB		25	170	600	550	660	24	6	660	527	336	163	140	430	460	269	1296	322	180
—	132 160	90 110	75	315MC		25	170	600	550	660	24	6	660	527	336	163	140	430	460	269	1296	322	180

Note : 1. Tolerance of Shaft End Diameter D : m6

2. Tolerance of Key Width F : h9

3. Tolerance of Shaft End Diameter D : m6

4. Tolerance of Key Width F : h9

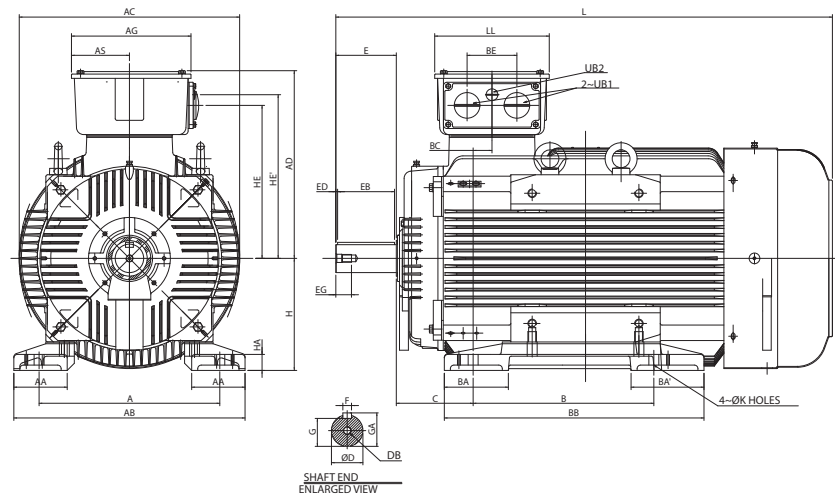


FIG. 9

Dimension in mm

UB1	UB2	SHAFT EXTENSION									BEARING	
		D	E	EB	ED	EG	F	G	GA	DB	DRIVE END	OPPOSITE DRIVE END
M63X1.5	M20X1.5	65	140	125	7.5	42	18	58	69	M20	6316C3	6314C3
M63X1.5	M20X1.5	75	140	125	7.5	42	20	67.5	79.5	M20	6318C3	6316C3
M63X1.5	M20X1.5	75	140	125	7.5	42	20	67.5	79.5	M20	NU318	6316C3
M63X1.5	M20X1.5	65	140	125	7.5	42	18	58	69	M20	6316C3	6314C3
M63X1.5	M20X1.5	75	140	125	7.5	42	20	67.5	79.5	M20	6318C3	6316C3
M63X1.5	M20X1.5	75	140	125	7.5	42	20	67.5	79.5	M20	NU318	6316C3
M63X1.5	M20X1.5	65	140	125	7.5	42	18	58	69	M20	6316C3	6314C3
M63X1.5	M20X1.5	80	170	160	5	42	22	71	85	M20	6320C3	6316C3
M63X1.5	M20X1.5	80	170	160	5	42	22	71	85	M20	NU320	6316C3
M63X1.5	M20X1.5	65	140	125	7.5	42	18	58	69	M20	6316C3	6314C3
M63X1.5	M20X1.5	80	170	160	5	42	22	71	85	M20	6320C3	6316C3
M63X1.5	M20X1.5	80	170	160	5	42	22	71	85	M20	NU320	6316C3

SPECIFICATION TABLE

IE3 EFFICIENCY 3-PHASE INDUCTION MOTORS

TYPE: AESV3E, AESU3E

ITEM		STANDARD SPECIFICATION
RATING	Kind of Motors	Squirrel - Cage Induction Motors (SCIM) .
	Design Standards	IEC 60034, IEC 60072 .
	Voltages	230V, 265V, 400V, 460V, 690V .
	Frequency	50Hz or 60Hz .
	Output Range	0.37 kW ~ 315 kW (50Hz) or 0.43 kW ~ 362 kW (60Hz) .
	R.P.M. (Syn.)	3000 ~ 750 R.P.M. (2 ~ 8 Poles) or 3600 ~ 900 R.P.M. (2 ~ 8 Poles) .
	Time Duty	Continuous. S1 , S.F. : 1.0 .
	Frame Nos.	80M ~ 315D .
	Protection Enclosure	Totally Enclosed (IP 55) .
	Cooling Method	Self External Fan, Surface Cooling (IC 411) .
	Mounting	Horizontal Foot Mounted B3 (IM 1001) . Horizontal Flange Mounted B5 (IM 3001) . Horizontal Foot And Flange Mounted B35 (IM 2001) . Vertical Flange Mounted , Shaft Down V1 (IM 3011) .
APPLICATION	Environment Conditions	Place : Shadow, Non-Hazardous. Ambient Temperature : -20 ~ 40°C . Relative Humidity : Less Than 90%RH (Non-Condensation) . Altitude : Less Than 1,000 Meters .
	Power Source Conditions	Voltage : $\pm 10\%$, Frequency : $\pm 5\%$, and 10% Max. of Combined Voltage and Frequency. But Frequency Variation Does Not Exceed $\pm 5\%$.
	Method of Starting	Full Voltage Direct On Line or Y - Δ Starting .
	Drive Method	Belt Service , However , 2 Pole 22 kW and Up Coupling Service is the Way .
	Direction of Rotation	Bi-Directional .
	Bearing	Bracket Mounting , Vacuum De - Gassed High Quality Open Bearings for Frame Nos. 180M ~ 315D, Grease Pre - Packed Shielded Rolling Bearings for the Others .
	Lead Terminal	Solderless Lug Terminals .
	Terminal Box	Cast iron, Can be Rotated Each 90° With Conduit Hole for Cable Entrance . The Terminal Box is Usually Placed on The Top of The Frame and The Cables Could be Let Out from Four Possible Directions .
	Stator Insulation	Class F Insulation System .
	Painting	Phenolic Rust Proof Base Plus Lacquer Surface Finished Painting in Pebble - Gray Color (Munsell 5Y 7.5/1) (RAL 7032) .
	Bolt Thread	ISO Metric System (Strength Catagory 8.8T) .
	Grounding Terminal	Two Terminals . One at inside of Terminal Box and Another One At Outside Of Frame .
PERFORMANCE	Test Procedure	IEC 60034-2-1 And Full Voltage Measuring Starting Performance .
	Winding Temperature Rise	Not to Exceed 80°C Rise by Resistance Method at S.F 1.0 Operation .
	Over Speed	120% Syn. R.P.M. for 2 Min .
	Over Torque	160% Rated Torque for 15 Sec .

ELECTRICAL CHARACTERISTICS - AESV3E / AESU3E

IE3

TEFC, CLASS F, 40°C AMBIENT TEMP., IEC DESIGN N CONTINUOUS DUTY, S.F. 1.0

400V/50HZ

OUTPUT		FULL LOAD rpm	FRAME NO.	EFFICIENCY				POWER FACTOR				CURRENT		TORQUE				ROTOR GD2 kg-m2	APPROX. WEIGHT kg
HP	kW			FULL LOAD (%)	3/4 LOAD (%)	2/4 LOAD (%)	1/4 LOAD (%)	FULL LOAD (%)	3/4 LOAD (%)	2/4 LOAD (%)	1/4 LOAD (%)	FULL LOAD (A)	LOCKED ROTOR (A)	FULL LOAD N·m	LOCKED ROTOR %FLT	PULL UP %FLT	BREAK DOWN %FLT		
0.5	0.37	920	80M	70.0	69.0	64.8	49.9	65.0	54.5	41.5	27.5	1.17	5	3.835	230	210	250	0.010	18.0
0.75	0.55	1430	80M	79.0	77.9	74.7	62.8	69.0	58.5	44.5	27.5	1.46	9	3.667	300	270	320	0.010	17.5
		905	80M	70.0	69.7	66.2	52.1	69.0	58.0	44.5	29.0	1.64	6	5.795	210	195	250	0.012	19.5
1	0.75	2875	80M	80.7	78.3	75.1	63.0	84.5	78.0	66.5	46.5	1.59	12	2.487	280	275	335	0.006	17.5
		1410	80M	82.5	81.8	79.7	71.6	73.5	64.0	50.0	31.5	1.79	11	5.072	315	290	335	0.013	20.5
		935	90S	78.9	80.6	79.4	72.3	71.0	62.5	49.0	30.5	1.93	9	7.649	210	190	250	0.022	27.5
1.5	1.1	2870	80M	82.7	83.0	81.3	73.8	85.0	78.5	66.5	45.0	2.26	18	3.655	300	295	350	0.007	19.5
		1430	90S	84.1	84.4	83.2	76.5	79.5	71.5	57.5	36.5	2.37	17	7.335	255	205	300	0.019	26.0
		930	90L	81.0	81.2	80.5	74.4	72.0	63.5	50.0	31.0	2.72	13	11.28	210	185	240	0.026	30.0
2	1.5	2850	90S	84.2	85.4	85.8	81.0	90.5	87.0	78.0	57.5	2.84	22	5.019	220	210	300	0.012	25.0
		1435	90L	85.3	84.1	82.2	74.3	75.0	65.5	51.5	31.5	3.38	26	9.967	300	235	335	0.023	28.0
		950	100L	82.5	82.9	81.5	73.9	72.5	65.0	52.0	33.0	3.62	18	15.06	210	175	250	0.058	41.0
3	2.2	2860	90L	85.9	86.7	86.8	82.2	89.5	85.0	75.5	53.5	4.13	35	7.335	245	235	315	0.014	28.0
		1450	100L	86.7	87.3	86.9	80.2	81.0	73.5	60.5	37.5	4.52	33	14.47	210	160	300	0.045	38.0
		960	112M	84.3	84.3	82.2	74.0	67.0	59.0	47.0	29.0	5.62	29	21.85	190	180	280	0.083	52.5
4	3	2855	100L	87.1	88.3	88.4	84.9	90.0	86.5	78.5	58.5	5.52	48	10.02	325	310	355	0.025	38.0
		1455	100L	87.7	87.7	86.2	78.9	78.0	70.5	57.5	36.5	6.33	49	19.66	250	240	335	0.052	40.5
		970	132S	85.6	86.1	85.1	79.9	79.5	73.0	60.0	38.0	6.36	41	29.49	195	170	300	0.154	74.0
5.5	4	2875	112M	88.1	89.0	88.9	85.5	91.0	87.5	80.0	61.0	7.20	69	13.27	270	250	360	0.046	50.0
		1445	112M	88.6	88.4	87.9	83.0	82.0	76.5	65.5	43.5	7.95	57	26.40	245	205	300	0.083	54.0
		970	132M	86.8	87.2	86.3	81.5	79.5	72.5	60.0	37.5	8.37	57	39.32	200	185	310	0.205	84.0
7.5	5.5	2930	132S	89.2	89.8	89.5	85.5	88.5	86.0	79.5	58.0	10.1	80	17.90	210	205	340	0.075	70.5
		1455	132S	89.6	90.4	90.3	86.9	85.5	81.0	70.5	49.0	10.4	77	36.04	240	200	300	0.132	74.5
		970	132M	88.0	88.0	86.2	79.4	73.5	65.5	52.0	31.5	12.3	88	54.07	210	205	345	0.216	87.0
10	7.5	2920	132S	90.1	90.9	90.8	87.7	87.0	84.5	77.5	58.5	13.8	100	24.49	210	195	315	0.081	75.0
		1460	132M	90.4	90.9	90.6	87.2	84.5	79.5	69.0	46.0	14.2	110	48.98	270	225	330	0.172	85.0
		970	160M	90.0	91.0	90.0	85.0	79.0	73.0	61.0	39.0	15.2	110	73.73	235	210	300	0.483	110
15	11	2935	160M	91.5	92.5	92.5	89.5	90.0	89.0	83.5	69.5	19.3	150	35.74	230	185	300	0.183	110
		1460	160M	91.4	92.0	91.5	88.0	84.0	80.0	70.0	49.0	20.7	160	71.84	230	185	300	0.366	133
		970	160L	90.3	91.0	90.5	86.0	78.0	72.0	60.5	38.0	22.5	170	108.1	295	255	300	0.628	138
20	15	2935	160M	91.9	92.0	92.0	88.0	89.0	85.5	77.5	57.5	26.5	230	48.73	275	230	330	0.205	120
		1460	160L	92.1	92.5	92.5	89.0	84.5	81.0	71.0	49.0	27.8	225	97.97	250	195	300	0.460	138
		970	180L	91.2	92.0	92.0	89.5	82.0	78.0	68.0	46.0	29.0	200	147.5	215	165	255	1.337	205
25	18.5	2930	160L	92.4	93.0	93.0	91.0	90.0	89.5	84.0	67.5	32.1	260	60.21	245	200	300	0.237	137
		1475	180M	92.6	94.0	93.0	90.0	81.5	77.0	69.5	46.0	35.4	270	119.6	215	160	280	0.704	183
		975	200L	92.0	93.0	93.0	90.0	80.5	76.0	66.5	44.0	36.1	260	180.9	220	185	265	1.829	263
30	22	2940	180M	93.0	93.0	93.0	89.0	87.0	85.0	77.0	53.0	39.2	300	71.35	225	180	300	0.283	178
		1475	180L	93.0	93.5	93.0	90.0	81.0	77.0	71.0	46.5	42.2	315	142.2	210	145	275	0.789	199
		975	200L	92.2	93.0	93.5	91.0	81.5	77.0	68.0	45.0	42.3	305	215.2	210	185	265	2.078	283

ELECTRICAL CHARACTERISTICS - AESV3E / AESU3E
TEFC, CLASS F, 40°C AMBIENT TEMP., IEC DESIGN N CONTINUOUS DUTY, S.F. 1.0
400V/50HZ

OUTPUT		FULL LOAD rpm	FRAME NO.	EFFICIENCY				POWER FACTOR				CURRENT		TORQUE				ROTOR GD2 kg-m2	APPROX. WEIGHT kg
HP	kW			FULL LOAD (%)	3/4 LOAD (%)	2/4 LOAD (%)	1/4 LOAD (%)	FULL LOAD (%)	3/4 LOAD (%)	2/4 LOAD (%)	1/4 LOAD (%)	FULL LOAD (A)	LOCKED ROTOR (A)	FULL LOAD N·m	LOCKED ROTOR %FLT	PULL UP %FLT	BREAK DOWN %FLT		
40	30	2950	200L	93.3	93.5	92.5	91.5	90.0	90.0	86.5	74.0	51.6	400	96.97	200	145	300	0.602	276
		1470	200L	94.1	95.0	95.0	93.0	86.0	84.5	77.0	56.0	53.5	445	194.6	250	205	300	1.451	266
		980	225M	93.0	94.0	94.0	91.0	83.5	80.0	76.5	53.0	55.8	335	291.9	210	160	240	3.023	343
50	37	2955	200L	94.0	95.0	94.5	92.5	91.0	90.5	87.0	75.0	62.4	510	119.4	210	145	300	0.753	302
		1480	225S	94.1	95.0	94.5	92.0	85.5	82.0	73.0	50.0	66.4	505	238.4	210	175	300	1.896	333
		980	250M	93.3	94.0	94.0	91.0	85.0	81.5	75.0	52.0	67.3	490	360.0	230	200	280	4.194	458
60	45	2960	225M	94.0	94.0	93.5	90.0	91.0	91.0	88.0	70.5	75.9	615	145.0	170	140	300	1.187	333
		1480	225M	94.5	95.0	94.5	92.0	84.5	79.5	70.0	47.0	81.3	600	289.9	210	175	300	1.979	368
		980	280S	93.7	93.7	93.0	90.5	85.0	82.0	73.5	52.0	81.6	530	438.5	140	125	230	7.600	675
75	55	2970	250M	95.0	95.0	95.0	92.0	91.5	90.0	86.5	72.0	91.3	735	176.6	165	130	315	1.544	456
		1485	250M	95.0	95.0	94.5	91.0	87.5	84.5	77.0	55.0	95.5	750	353.2	210	185	295	3.911	492
		980	280M	94.1	94.0	93.5	90.5	85.0	82.0	74.0	52.0	99.3	640	536.0	150	125	240	9.600	745
100	75	2965	280S	94.7	94.5	93.0	90.0	91.0	90.0	85.5	68.0	125.6	910	241.6	150	130	240	2.400	705
		1480	280S	95.0	95.0	94.2	91.8	89.0	87.5	82.0	63.0	128.0	910	482.6	160	135	250	6.400	710
		985	315S	94.6	94.5	93.7	90.5	84.5	81.5	73.0	51.0	135.4	900	727.2	160	135	240	14.80	985
125	90	2970	280M	95.0	95.0	94.6	91.5	90.5	89.0	83.0	63.5	151.1	1100	289.4	150	135	240	2.800	740
		1480	280M	95.2	95.0	94.5	91.5	88.5	87.0	81.0	61.5	154.2	1100	580.7	180	150	250	7.200	755
		985	315M	94.9	94.9	94.2	91.5	85.0	82.5	75.0	53.0	161.0	1030	872.6	160	130	230	16.80	1,055
150	110	2970	315S	95.2	95.0	94.3	90.5	90.5	89.0	85.0	67.5	184.3	1335	353.7	150	125	240	4.800	970
		1480	315S	95.4	95.2	94.8	92.0	88.5	86.5	81.0	60.5	188.1	1360	709.8	160	135	250	11.60	1,080
		985	315M	95.1	95.0	94.5	91.5	84.0	80.0	70.5	48.5	198.8	1380	1066	170	140	250	20.40	1,170
175	132	2970	315M	95.4	95.2	94.8	92.0	91.0	90.5	88.0	74.0	219.5	1500	424.4	130	120	230	5.200	1,025
		1480	315M	95.6	95.5	94.8	92.0	88.5	87.5	85.0	70.0	225.2	1500	851.8	150	125	230	12.40	1,100
		985	315L	95.4	95.4	95.0	92.8	87.0	85.5	78.5	57.2	229.6	1500	1280	130	140	230	21.20	1,345
215	160	2975	315M	95.6	95.5	94.8	92.0	92.0	91.0	87.5	72.0	262.6	1890	513.6	130	125	240	5.200	1,075
		1482	315M	95.8	95.8	95.0	92.5	90.0	89.0	83.5	65.0	267.9	1850	1031	125	105	230	12.00	1,220
		985	315L	95.6	95.6	95.0	93.8	87.0	85.0	78.0	56.5	277.7	1860	1551	130	110	230	23.20	1,435
270	200	2975	315L	95.8	95.8	95.5	92.6	92.0	91.5	90.0	77.0	327.5	2320	642.0	130	100	240	6.800	1,270
		1482	315L	96.0	95.8	95.3	93.2	90.0	89.5	85.0	67.0	334.1	2270	1289	130	105	230	15.20	1,400
		985	315D	95.8	95.8	95.4	93.5	88.0	86.5	80.0	60.0	342.4	2450	1939	150	130	240	43.60	1,930
335	250	2975	315D	95.8	95.8	94.8	91.5	91.5	91.0	88.5	76.0	411.7	2850	802.5	140	120	250	10.40	1,765
		1485	315D	96.0	95.8	95.0	92.6	91.0	90.0	88.0	75.0	413.1	2950	1608	130	125	240	29.20	1,990
		985	315D	95.8	96.0	95.5	94.2	88.0	86.0	80.5	60.0	428.0	3040	2424	150	130	230	46.00	2,010
420	315	2970	315D	95.8	95.5	94.7	91.5	92.0	91.5	90.5	80.0	515.9	3760	1013	130	105	240	11.60	1,885
		1485	315D	96.0	96.0	95.5	93.5	91.5	91.0	89.0	75.5	517.6	3800	2026	120	100	240	34.00	2,110

- Note :
1. The above are typical values based on test according to IEC 60034-2-1.
 2. Tolerance according to IEC 60034-1.
 3. Efficiency, power factor, speed and torque are the same for other voltages.
Current values vary inversely with voltage.
 4. 0.55 kW and below : Efficiency per TECO performance standard.
 5. Data subject to change without notice.

ELECTRICAL CHARACTERISTICS - AESV3E / AESU3E

IE3

TEFC, CLASS F, 40°C AMBIENT TEMP., IEC DESIGN N CONTINUOUS DUTY, S.F. 1.0
S.F. 1.0 50HZ 400V WINDING USED ON SF1.15 60HZ 460V

460V

OUTPUT		FULL LOAD rpm	FRAME NO.	EFFICIENCY				POWER FACTOR				CURRENT		TORQUE				ROTOR GD2 kg-m2	APPROX. WEIGHT kg
HP	kW			FULL LOAD (%)	3/4 LOAD (%)	2/4 LOAD (%)	1/4 LOAD (%)	FULL LOAD (%)	3/4 LOAD (%)	2/4 LOAD (%)	1/4 LOAD (%)	FULL LOAD (A)	LOCKED ROTOR (A)	FULL LOAD N·m	LOCKED ROTOR %FLT	PULL UP %FLT	BREAK DOWN %FLT		
0.58	0.43	1120	80M	76.0	76.4	73.1	59.5	64.5	55.0	42.5	27.0	1.10	5	3.667	225	220	255	0.010	18.0
0.84	0.63	1725	80M	80.0	78.8	74.3	61.3	70.0	62.5	48.0	30.5	1.41	9	3.488	325	295	355	0.010	17.5
		1110	80M	76.0	76.4	73.5	60.2	66.5	57.0	44.0	28.0	1.56	7	5.420	210	205	240	0.012	19.5
1.15	0.86	3480	80M	81.6	80.5	76.7	65.1	84.0	78.0	67.0	48.0	1.57	14	2.360	295	285	370	0.006	17.5
		1710	80M	84.0	83.6	81.4	73.6	74.0	65.0	51.0	32.0	1.74	12	4.803	370	340	355	0.013	20.5
		1140	90S	82.5	83.8	82.7	73.0	70.5	62.0	48.5	30.0	1.86	10	7.204	195	170	250	0.022	27.5
1.7	1.27	3470	80M	84.0	83.9	81.6	72.9	85.5	79.5	68.0	47.0	2.22	19	3.495	320	310	365	0.007	19.5
		1730	90S	85.5	86.1	84.6	78.0	78.0	70.5	57.5	37.5	2.39	17	7.011	250	195	300	0.019	26.0
		1130	90L	82.5	83.3	82.6	77.3	72.0	64.0	50.5	31.5	2.68	13	10.73	185	165	235	0.026	30.0
2.32	1.73	3455	90S	85.5	86.7	86.5	81.5	90.0	86.0	77.5	58.0	2.82	23	4.782	295	210	300	0.012	25.0
		1735	90L	85.5	85.7	83.9	76.9	75.5	67.0	53.0	33.0	3.36	25	9.522	280	220	330	0.023	28.0
		1150	100L	84.0	84.2	82.7	75.5	73.0	65.5	53.0	34.0	3.54	18	14.37	200	155	240	0.058	41.0
3.39	2.53	3460	90L	86.5	87.4	87.0	82.3	89.5	85.5	76.5	55.0	4.10	38	6.983	320	235	315	0.014	28.0
		1760	100L	88.5	88.1	86.9	79.2	74.5	65.5	52.0	32.0	4.82	41	13.73	245	195	350	0.045	38.0
		1155	112M	86.5	86.5	84.7	77.5	67.5	60.0	48.0	30.0	5.44	28	20.92	165	160	270	0.083	52.5
4.62	3.45	3450	100L	88.5	89.4	89.2	84.6	90.0	87.5	80.0	51.5	5.44	47	9.550	295	285	335	0.025	38.0
		1750	100L	88.5	88.5	87.1	80.7	79.5	72.5	60.0	38.5	6.15	50	18.83	225	220	340	0.052	40.5
		1165	132S	89.5	89.5	88.1	82.7	77.0	70.0	57.5	37.0	6.28	41	28.28	190	165	300	0.154	74.0
6.17	4.6	3470	112M	89.5	90.2	89.9	85.7	91.5	88.5	81.5	62.5	7.05	68	12.66	255	250	325	0.046	50.0
		1740	112M	88.5	89.2	88.8	84.3	82.0	77.5	67.0	45.5	7.96	57	25.25	220	175	300	0.083	54.0
		1170	132M	90.2	90.4	88.9	83.8	77.5	70.5	57.5	37.0	8.26	61	37.55	210	170	300	0.205	84.0
8.49	6.33	3530	132S	89.5	89.7	88.9	84.2	87.5	85.0	78.5	60.0	10.1	79	17.13	210	205	300	0.075	70.5
		1755	132S	91.7	92.1	91.9	88.0	85.5	81.0	71.0	48.5	10.1	78	34.45	255	205	330	0.132	74.5
		1170	132M	90.2	90.0	88.6	82.6	74.5	67.0	53.5	33.0	11.8	87	51.67	210	185	330	0.216	87.0
11.5	8.6	3520	132S	90.2	90.7	90.2	86.8	87.5	85.5	79.0	62.0	13.7	100	23.33	210	200	300	0.081	75.0
		1760	132M	91.7	92.2	91.9	88.7	85.5	81.0	71.0	49.0	13.8	113	46.66	285	235	350	0.172	85.0
		1170	160M	91.0	91.5	91.0	86.4	80.0	75.0	64.5	42.5	14.8	110	70.20	235	210	300	0.483	110
17	12.7	3525	160M	91.0	91.0	90.1	85.0	92.0	91.0	87.5	75.0	19.0	155	34.41	215	190	300	0.183	110
		1760	160M	92.4	93.0	92.9	90.1	87.0	84.0	76.0	55.0	19.8	165	68.91	255	240	355	0.366	133
		1170	160L	91.7	92.0	91.7	87.4	81.0	76.0	65.5	42.0	21.5	171	103.7	245	220	300	0.628	138
23	17.3	3535	160M	91.7	92.0	91.5	87.3	91.0	89.0	84.0	66.0	26.0	225	46.74	305	255	340	0.205	120
		1755	160L	93.0	93.5	93.4	90.9	87.0	84.0	76.0	55.0	26.8	225	94.14	285	280	385	0.460	138
		1165	180L	91.7	92.3	92.6	89.7	83.5	80.5	73.0	51.0	28.4	205	141.8	225	200	280	1.337	205
28.5	21.3	3525	160L	91.7	92.4	92.3	89.0	93.0	90.5	86.5	72.0	31.3	265	57.71	255	210	300	0.237	137
		1770	180M	93.6	94.0	93.5	90.1	82.5	79.0	71.0	49.5	34.6	265	114.9	230	225	355	0.704	183
		1175	200L	93.0	93.4	93.3	90.0	82.0	78.0	69.5	48.0	35.1	270	173.1	255	195	260	1.829	263
34	25.3	3540	180M	91.7	92.0	91.3	87.7	90.5	86.0	79.0	60.0	38.3	311	68.25	250	200	300	0.283	178
		1770	180L	93.6	94.0	93.5	90.3	82.0	79.0	71.0	49.0	41.4	310	136.5	235	230	355	0.789	199
		1175	200L	93.0	93.5	93.5	90.5	82.0	79.0	71.0	49.0	41.6	320	205.6	245	190	250	2.078	283

ELECTRICAL CHARACTERISTICS - AESV3E / AESU3E
TEFC, CLASS F, 40°C AMBIENT TEMP., IEC DESIGN N CONTINUOUS DUTY, S.F. 1.0
S.F. 1.0 50HZ 400V WINDING USED ON SF1.15 60HZ 460V
460V

OUTPUT		FULL LOAD rpm	FRAME NO.	EFFICIENCY				POWER FACTOR				CURRENT		TORQUE				ROTOR GD2 kg-m2	APPROX. WEIGHT kg
HP	kW			FULL LOAD (%)	3/4 LOAD (%)	2/4 LOAD (%)	1/4 LOAD (%)	FULL LOAD (%)	3/4 LOAD (%)	2/4 LOAD (%)	1/4 LOAD (%)	FULL LOAD (A)	LOCKED ROTOR (A)	FULL LOAD N-m	LOCKED ROTOR %FLT	PULL UP %FLT	BREAK DOWN %FLT		
46	34.5	3545	200L	93.0	92.9	92.0	87.5	91.0	90.5	88.0	78.0	51.2	400	92.94	200	160	220	0.602	276
		1770	200L	94.1*	94.5	94.3	92.3	89.0	87.0	81.5	64.5	51.7	410	186.1	210	195	300	1.451	266
		1175	225M	93.0*	93.7	93.7	90.6	87.0	86.0	81.0	61.0	53.5	340	280.4	175	120	210	3.023	343
57	42.6	3550	200L	93.6	93.8	93.6	90.2	91.0	91.0	88.0	78.0	62.8	510	114.6	205	150	255	0.753	302
		1780	225S	94.5*	94.7	94.1	90.4	86.0	83.0	75.5	54.0	65.8	510	228.6	210	190	310	1.896	333
		1180	250M	94.1*	94.3	94.0	90.7	87.5	86.0	80.0	59.0	64.9	500	344.8	220	195	250	4.194	458
69.5	51.8	3560	225M	93.6	93.6	92.8	88.4	93.0	93.0	92.5	80.0	74.7	620	139.0	180	175	300	1.187	333
		1775	225M	94.5*	94.5	94.1	90.9	86.0	83.5	76.5	55.5	80.0	620	278.7	235	200	295	1.979	368
		1185	280S	94.5	94.5	93.5	90.5	85.0	83.0	76.0	55.0	80.9	500	417.5	150	125	260	7.600	675
84.5	63	3565	250M	93.6	93.9	93.3	90.8	92.5	92.0	88.5	75.5	91.3	740	168.8	210	160	300	1.544	456
		1780	250M	95.0*	95.0	94.2	90.6	88.0	85.5	78.5	57.0	94.6	755	338.0	240	205	300	3.911	492
		1185	280M	94.6	94.5	93.5	90.6	85.2	83.0	76.0	55.0	98.1	632	507.7	160	135	270	9.600	745
115	86	3570	280S	94.5*	94.5	93.8	90.8	91.0	90.0	86.0	69.0	125.5	870	230.1	120	100	300	2.400	705
		1780	280S	95.4	95.4	94.2	91.0	89.5	88.5	84.0	67.0	126.4	870	461.4	170	145	260	6.400	710
		1185	315S	95.0	95.0	93.6	90.8	85.5	83.0	76.5	55.5	132.9	900	693.1	180	155	260	14.80	985
140	104	3570	280M	95.0	95.0	94.4	91.0	91.0	90.0	85.0	68.0	151.0	1130	278.2	140	115	310	2.800	740
		1780	280M	95.4*	95.4	94.5	91.5	89.5	88.5	84.0	67.0	152.9	1100	558.0	180	155	260	7.200	755
		1185	315M	95.2*	95.0	94.0	91.0	85.5	84.0	77.5	57.0	160.4	1100	838.1	180	155	240	16.80	1,055
170	127	3575	315S	95.2	95.2	94.5	91.0	91.0	90.0	85.5	69.0	184.0	1340	339.3	130	105	300	4.800	970
		1785	315S	95.8	95.7	94.5	91.5	89.5	88.5	82.5	64.5	185.9	1390	679.5	180	155	280	11.60	1,080
		1188	315M	95.8	95.8	94.2	91.5	85.5	83.0	76.0	54.0	194.6	1390	1021	210	185	270	20.40	1,170
204	152	3575	315M	95.6	95.5	94.8	91.5	91.5	91.0	88.0	74.0	218.1	1490	406	130	105	280	5.200	1,025
		1783	315M	95.8*	95.5	94.5	91.5	89.5	89.0	86.0	72.0	222.5	1385	814	150	125	230	12.40	1,100
		1188	315L	95.8	95.8	94.5	91.5	87.5	85.5	80.0	60.5	227.6	1480	1222	110	115	230	21.20	1,345
247	184	3577	315M	95.8	95.6	94.5	91.5	92.0	91.5	88.5	75.0	262.0	1835	491	110	115	270	5.200	1,075
		1785	315M	96.2	96.0	94.6	92.0	90.5	90.0	86.0	70.0	265.3	1840	984	110	115	240	12.00	1,220
		1188	315L	95.9	96.0	94.6	92.0	87.5	86.0	80.0	60.5	275.2	1805	1479	110	115	230	23.20	1,435
308	230	3578	315L	95.8	95.6	94.5	91.5	92.5	92.0	90.5	80.0	325.8	2250	613.9	110	115	250	6.800	1,270
		1788	315L	96.2	96.0	94.8	92.0	91.0	90.5	87.0	72.0	329.8	2210	1228	110	115	230	15.20	1,400
		1188	315D	96.0	96.0	94.8	92.0	88.0	86.5	81.0	61.0	341.7	2435	1849	130	138	240	43.60	1,930
385	288	3578	315D	95.8	95.6	94.5	91.5	92.5	92.0	91.0	81.0	407.9	2950	768.7	110	115	260	10.40	1,765
		1788	315D	96.2	96.0	95.0	92.2	91.5	91.0	89.0	75.0	410.7	3060	1538	110	115	240	29.20	1,990
		1188	315D	96.0	96.0	95.0	92.2	88.0	87.0	81.5	62.0	427.9	3060	2315	120	126	230	46.00	2,010
485	362	3578	315D	96.0	95.8	94.6	91.5	93.0	92.5	91.5	82.0	508.9	3740	966	110	115	260	11.60	1,885
		1788	315D	96.0*	95.8	95.0	92.2	92.0	91.5	89.5	77.0	514.5	3835	1934	110	115	250	34.00	2,110

- Note :
1. The above are typical values based on test according to IEC 60034-2-1.
 2. Tolerance according to IEC 60034-1.
 3. Efficiency, power factor, speed and torque are the same for other voltages.
Current values vary inversely with voltage.
 4. (*) Efficiency, per TECO performance standard (Not IE3).
 5. 0.63 kW and below : Efficiency per TECO performance standard.
 6. Data subject to change without notice.

MEMO

[illegible]